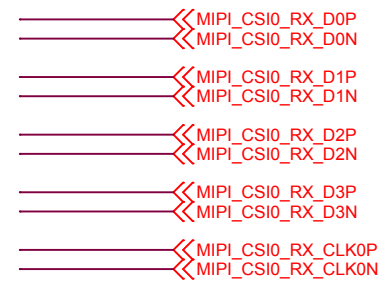
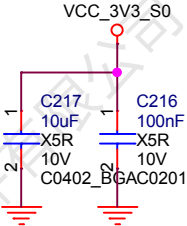
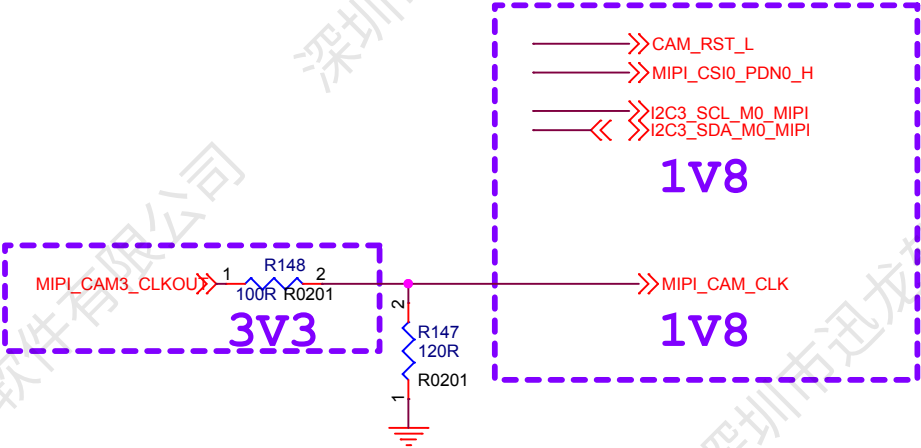
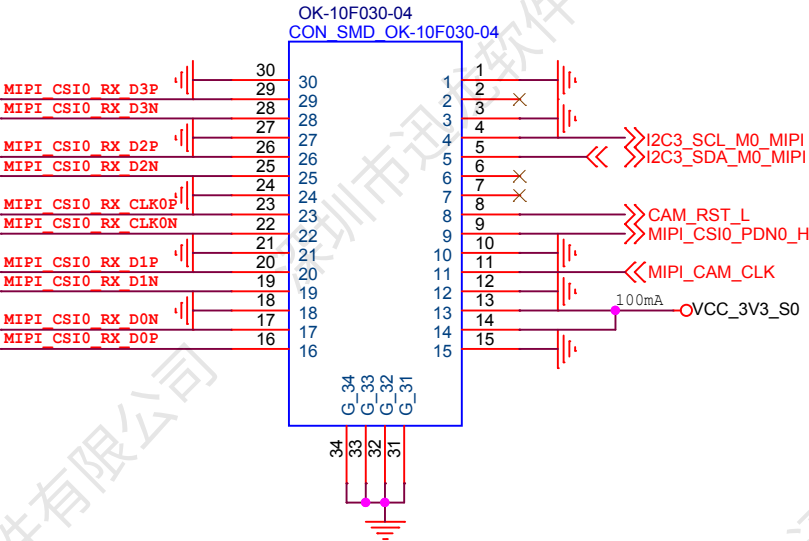


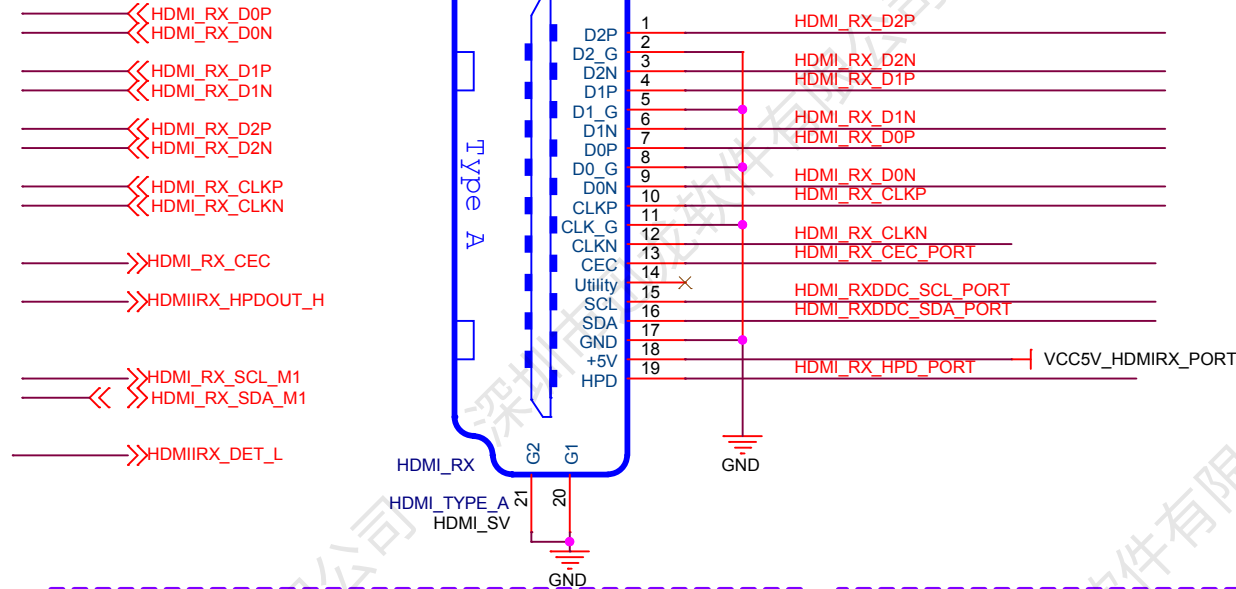
MIPI-CSIO_RX



CAM

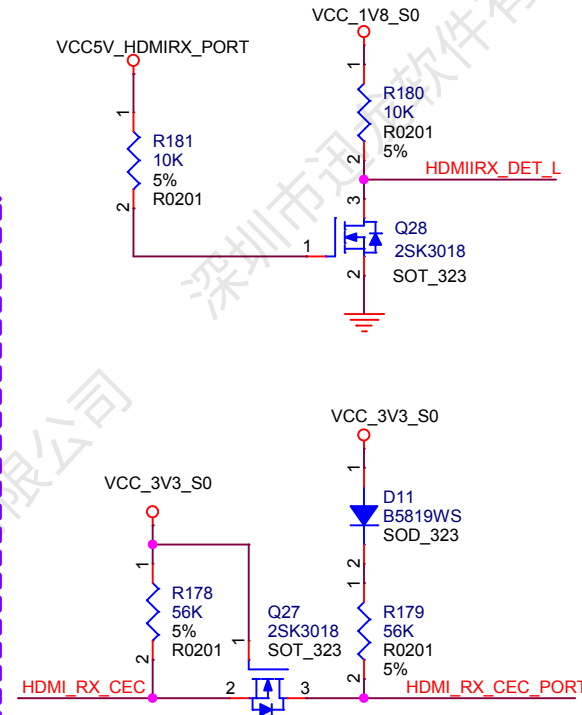
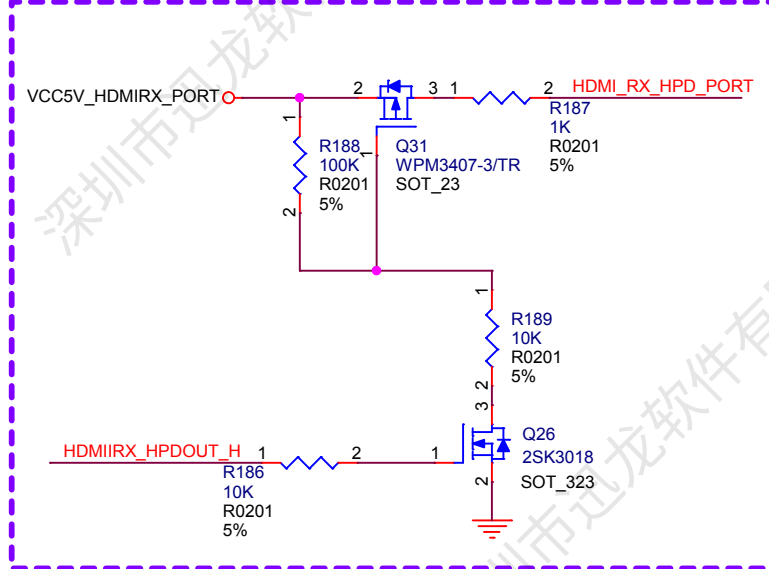
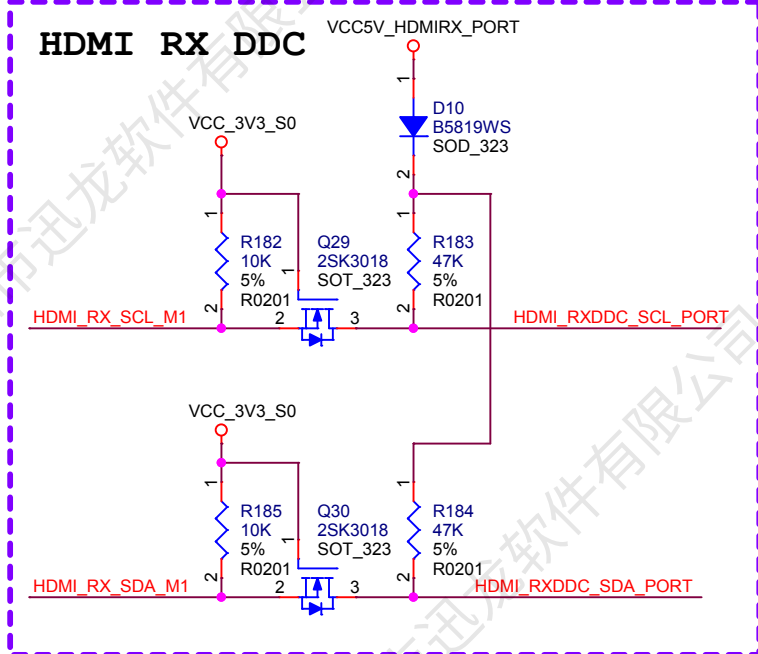


HDMI2.0 RX



Note:

(This HDMI2.0 RX interface has low probability anomalies, such as splash screen and restart, in scenarios such as frequent plug and pull, high and low resolution switching, etc. The probability is between 0.1% and 1%. We are still trying to debug and solve these problems. If the user will frequently use various peripherals to insert this HDMI2.0 RX interface, it is recommended to use external bridges connecting chips to improve compatibility to avoid affecting product production.)



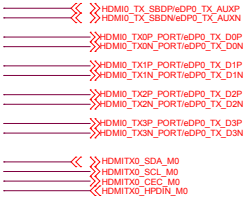
HPD:
Sink Side: Require output, Min 2.4V; Max 5.3V
Source Side: Require input and Detection.
Min 2.0V, Max: 5.3V



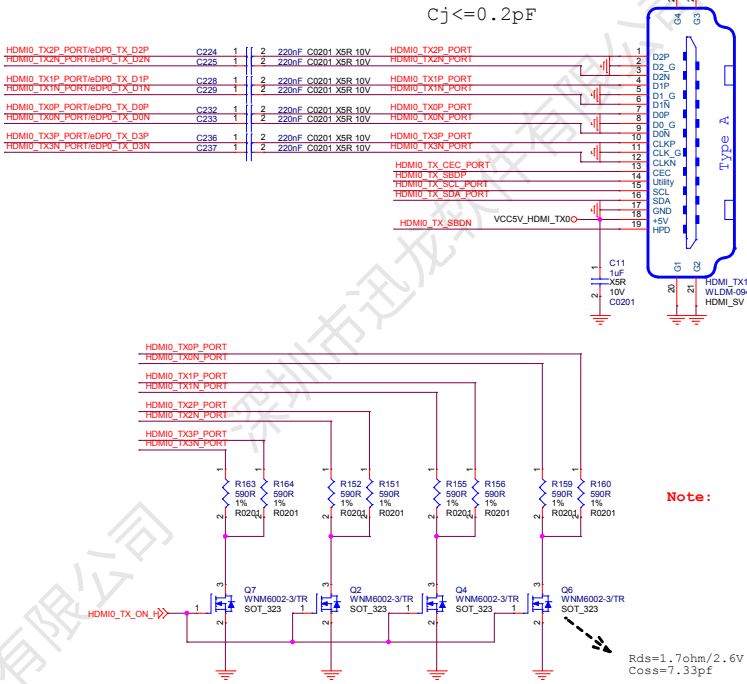
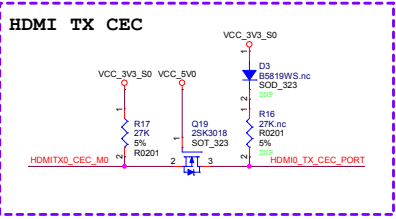
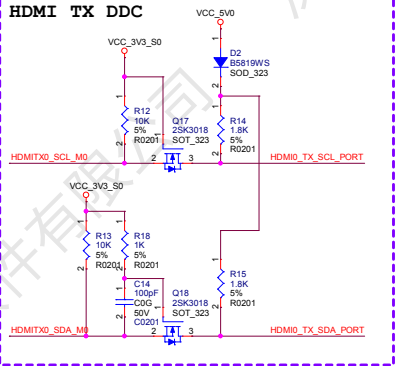
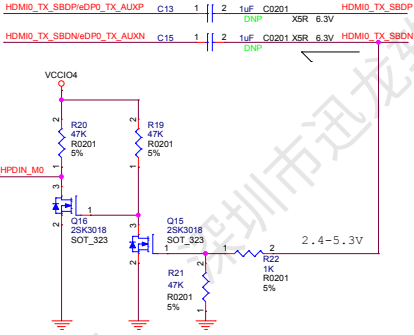
Shenzhen Xunlong Software Co., Ltd

Project:	OPI 5 PLUS		
File:	02.VI-HDMI2.0 RX		
Date:	Friday, July 25, 2025	Rev:	V1.3.1
Size:	A3	Sheet:	2
Of:	33		

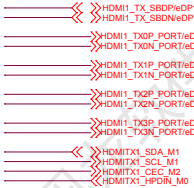
HDMI TX0



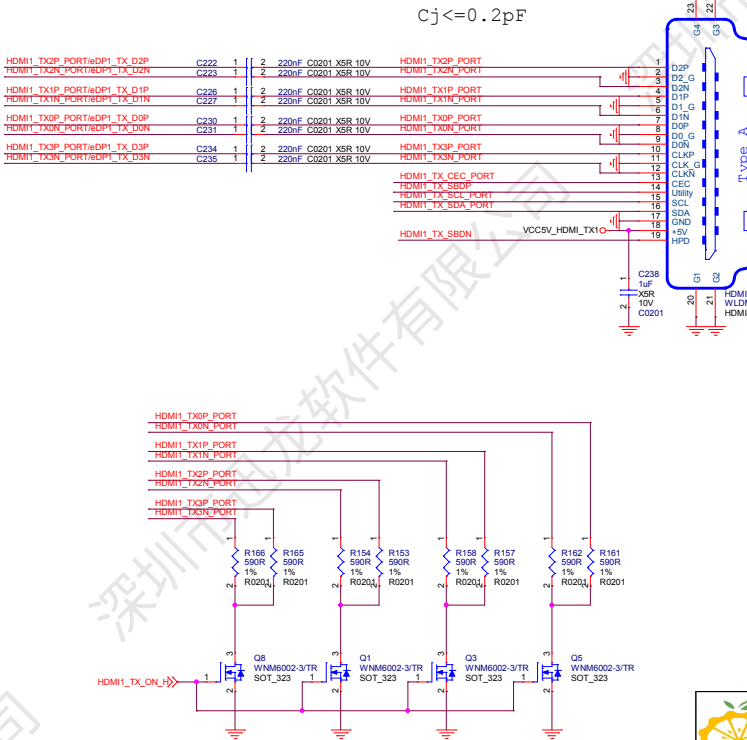
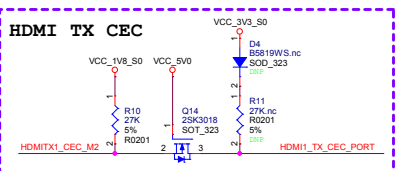
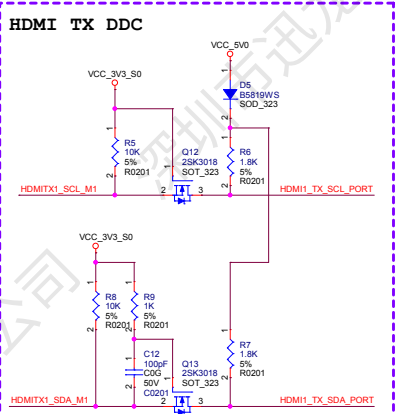
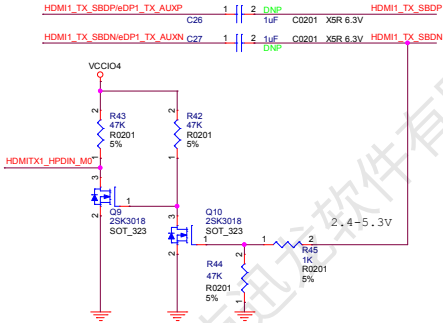
HDMI TX eARC



HDMI TX1

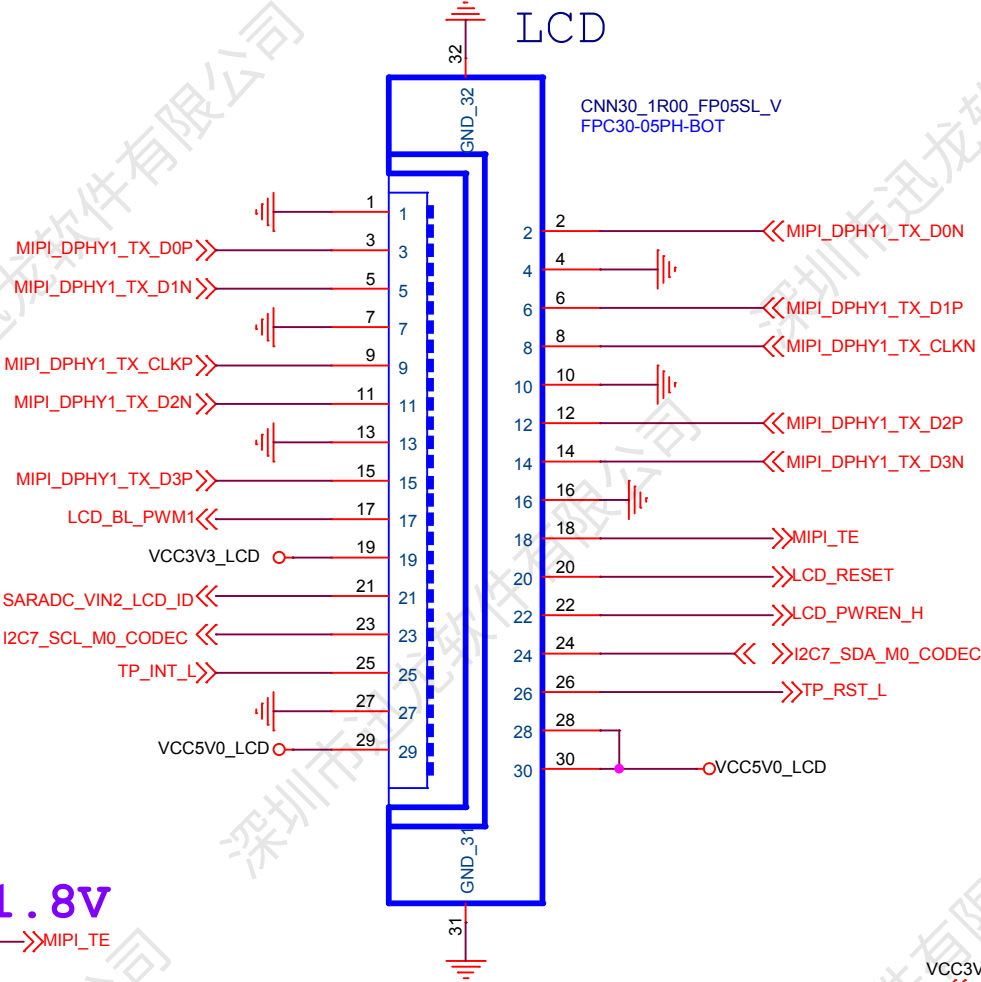
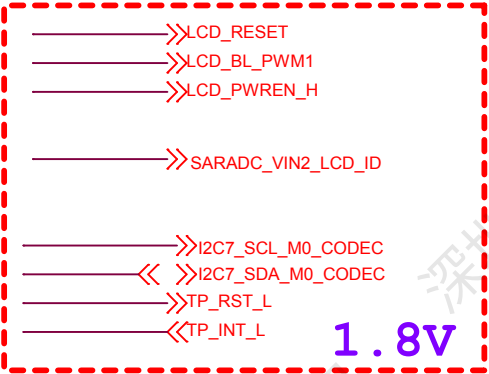


HDMI TX eARC



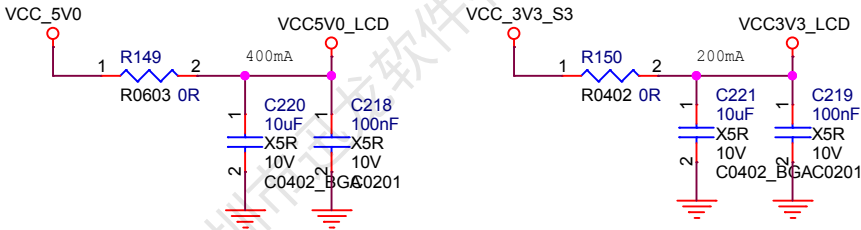
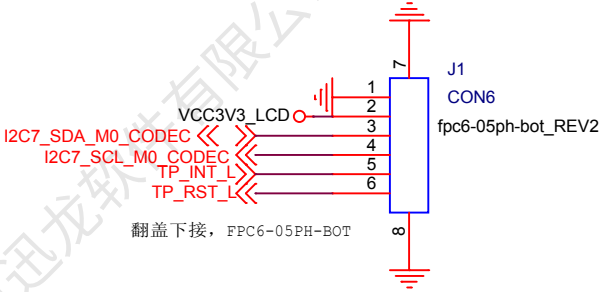
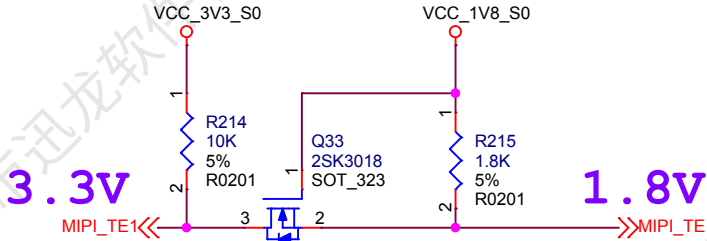
MIPI DPHY1 TX

MIPI DPHY1 TX



FPC Pin List

- Pin1 :GND
Pin2 :DON
Pin3 :DOP
Pin4 :GND
Pin5 :DIN
Pin6 :DIP
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0



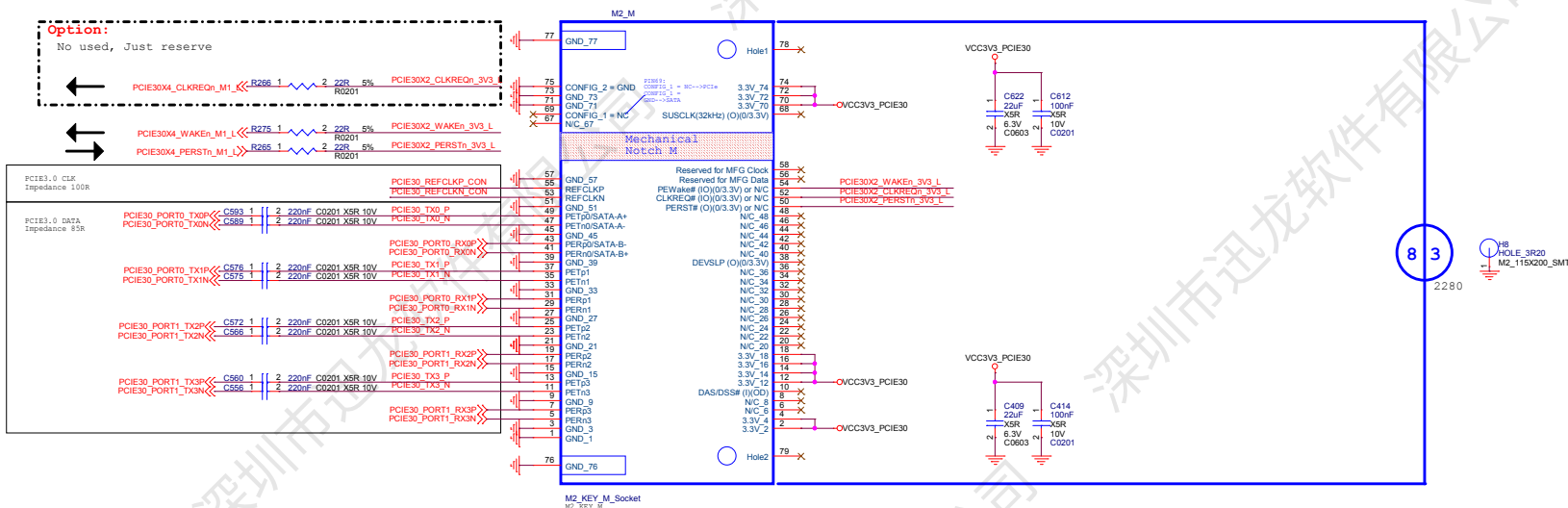
Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	04.VO-LCM_MIPI			
Date:	Friday, July 25, 2025	Rev:	V1.3.1	
Size:	A3	Sheet:	4	Of: 33

M.2_PCIe3.0 x 2Lanes

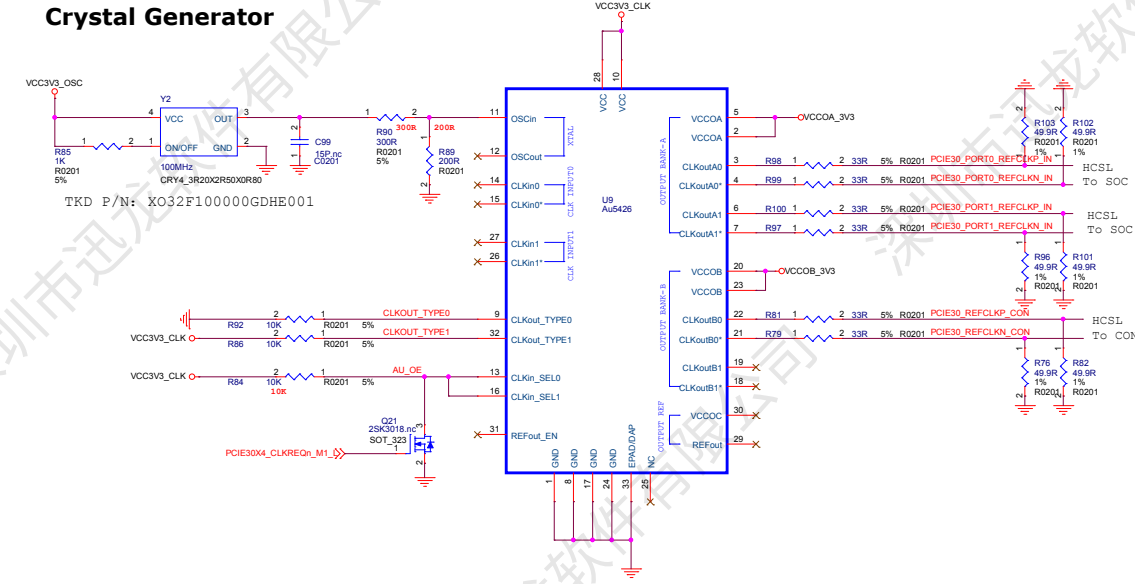
PCIE30_PORT0_REFCLKP_IN
PCIE30_PORT0_REFCLKN_IN
PCIE30_PORT0_TX0P
PCIE30_PORT0_TX0N
PCIE30_PORT0_TX1P
PCIE30_PORT0_TX1N
PCIE30_PORT0_RX0P
PCIE30_PORT0_RX0N
PCIE30_PORT0_RX1P
PCIE30_PORT0_RX1N

PCIE30_PORT1_REFCLKP_IN
PCIE30_PORT1_REFCLKN_IN
PCIE30_PORT1_TX2P
PCIE30_PORT1_TX2N
PCIE30_PORT1_TX3P
PCIE30_PORT1_TX3N
PCIE30_PORT1_RX2P
PCIE30_PORT1_RX2N
PCIE30_PORT1_RX3P
PCIE30_PORT1_RX3N

PCIE30X4_WAKEN_M1_L
PCIE30X4_PERSTn_M1_L
PCIE_PWREN_H



Crystal Generator



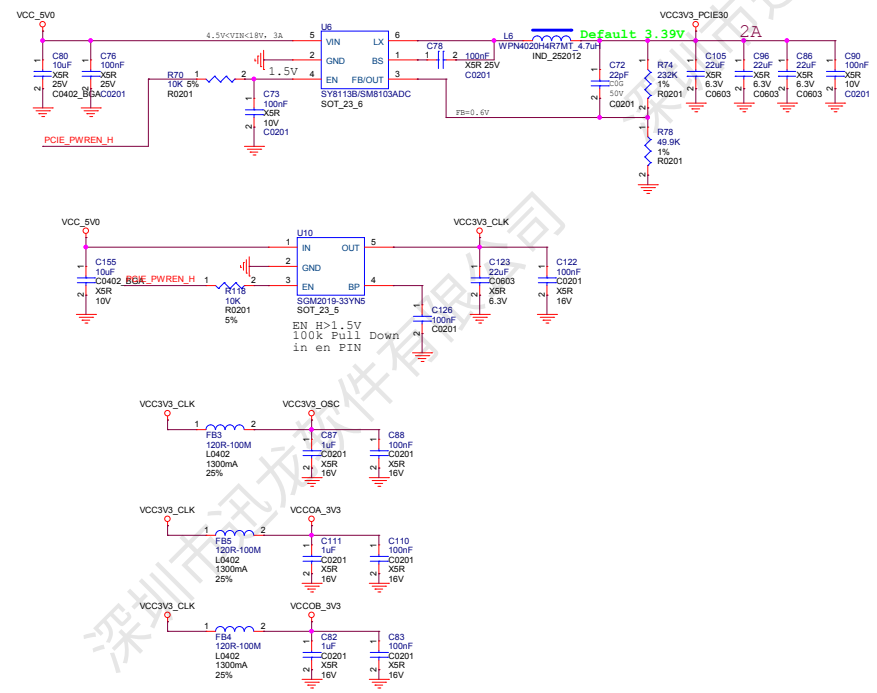
CLKIN SELECT

CLKIN_SEL1	CLKIN_SEL0	CLK Buffer Type
0	0	CLKIN0
0	1	CLKIN1
1	0	CRYSTAL_HF_40_MODE
1	1	CRYSTAL_LF_40_MODE

CLKOUT TYPE SELECT

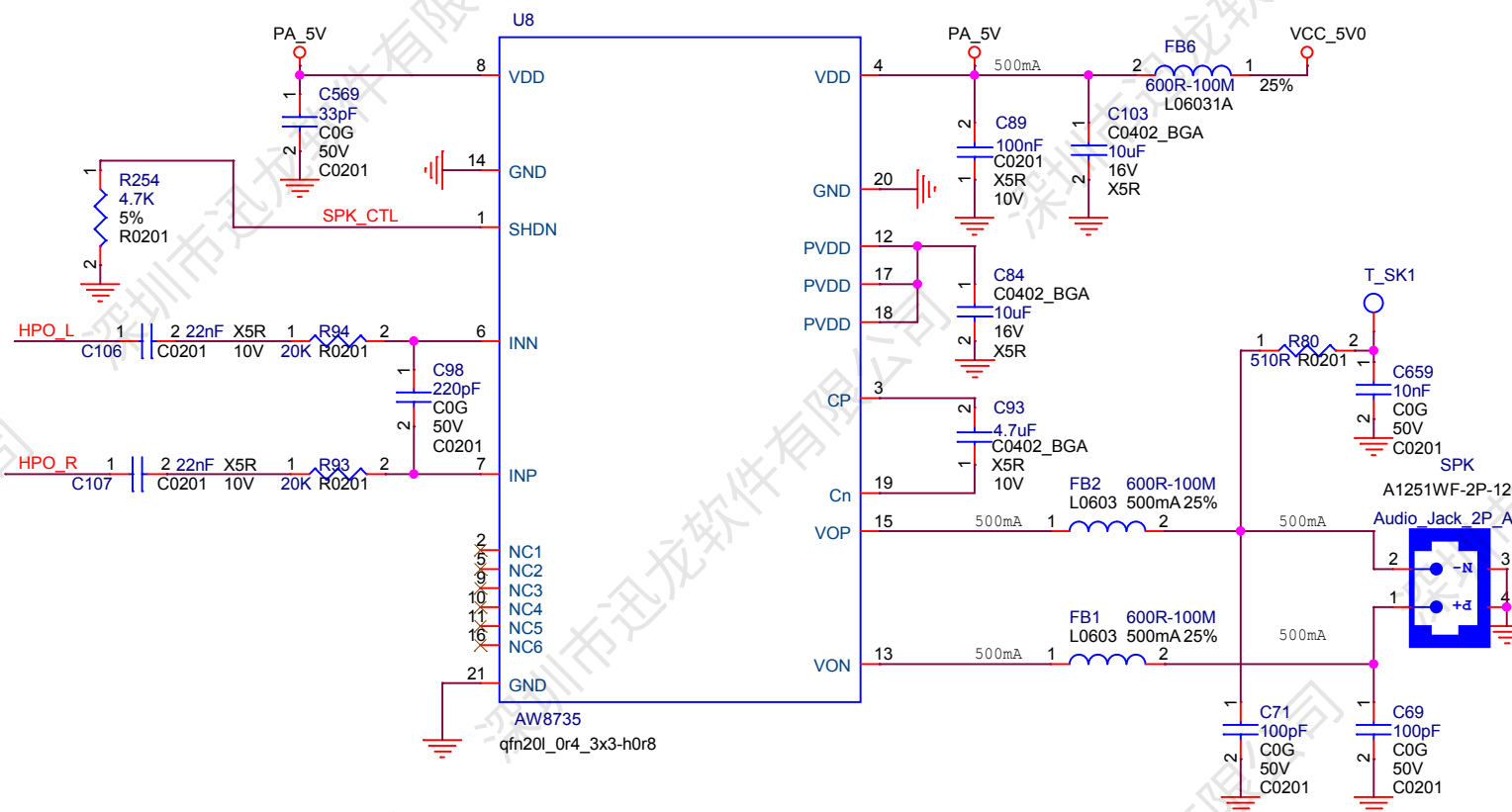
CLKOUT_TYPE1	CLKOUT_TYPE0	CLK Buffer Type
0	0	TYPE0
0	1	TYPE1
1	0	TYPE2
1	1	TYPE3

M.2 Power



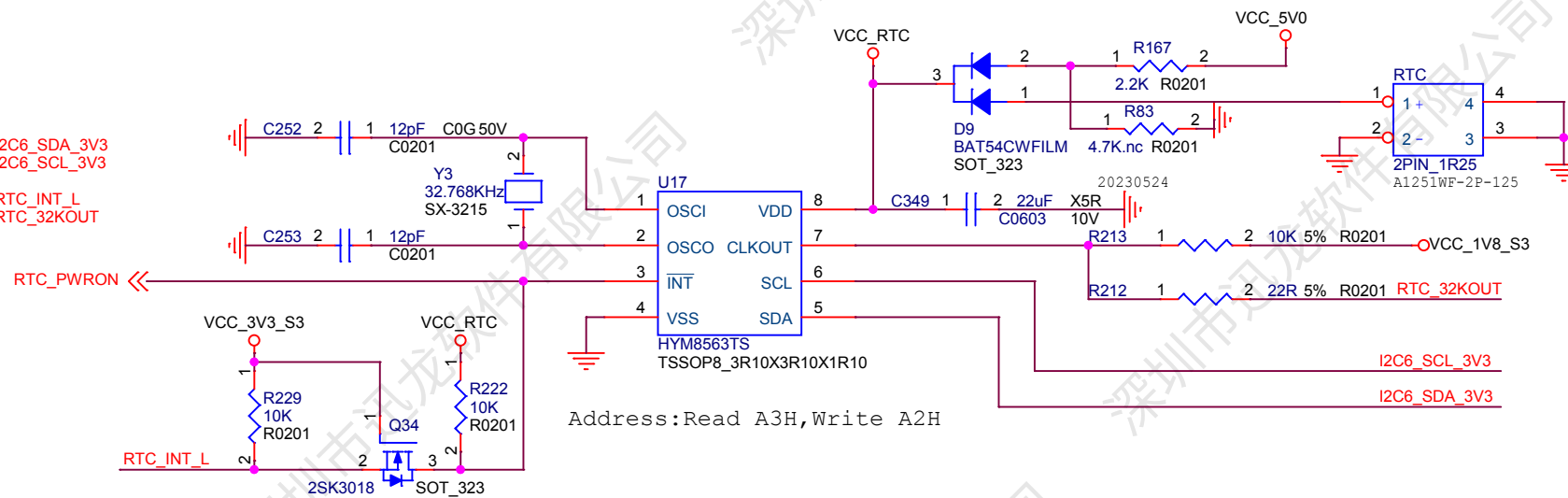
SPK PA

>>HPO_L
>>HPO_R
>>SPK_CTL

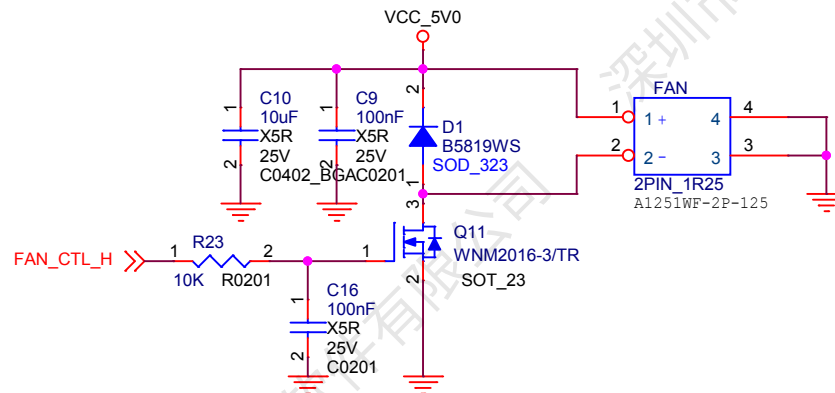


Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	07.SPK PA			
Date:	Friday, July 25, 2025			Rev: V1.3.1
Size:	A3	Sheet:	7	Of: 33

RTC IC



FAN_POWER



Shenzhen Xunlong Software Co., Ltd

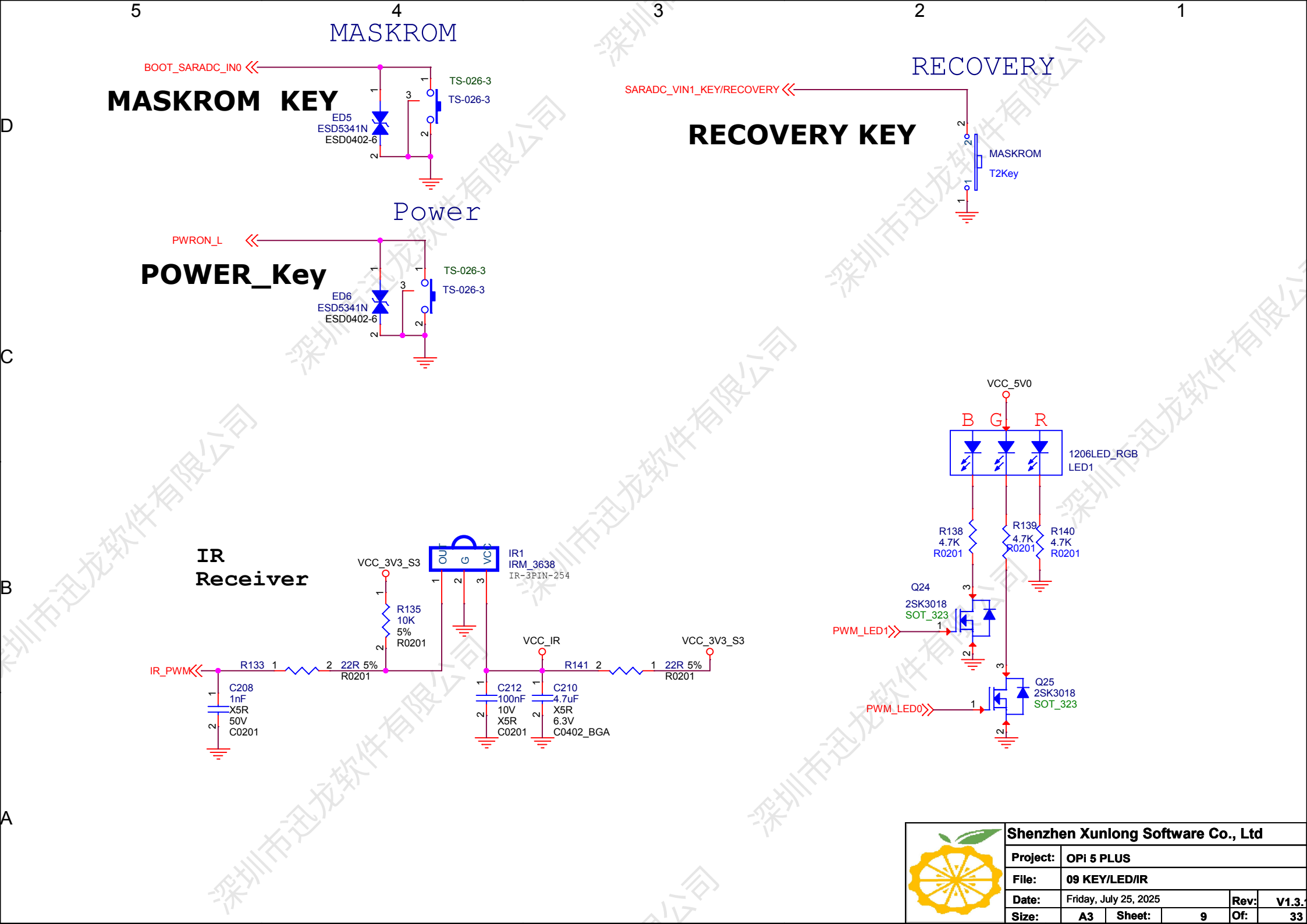
Project:	OPI 5 PLUS
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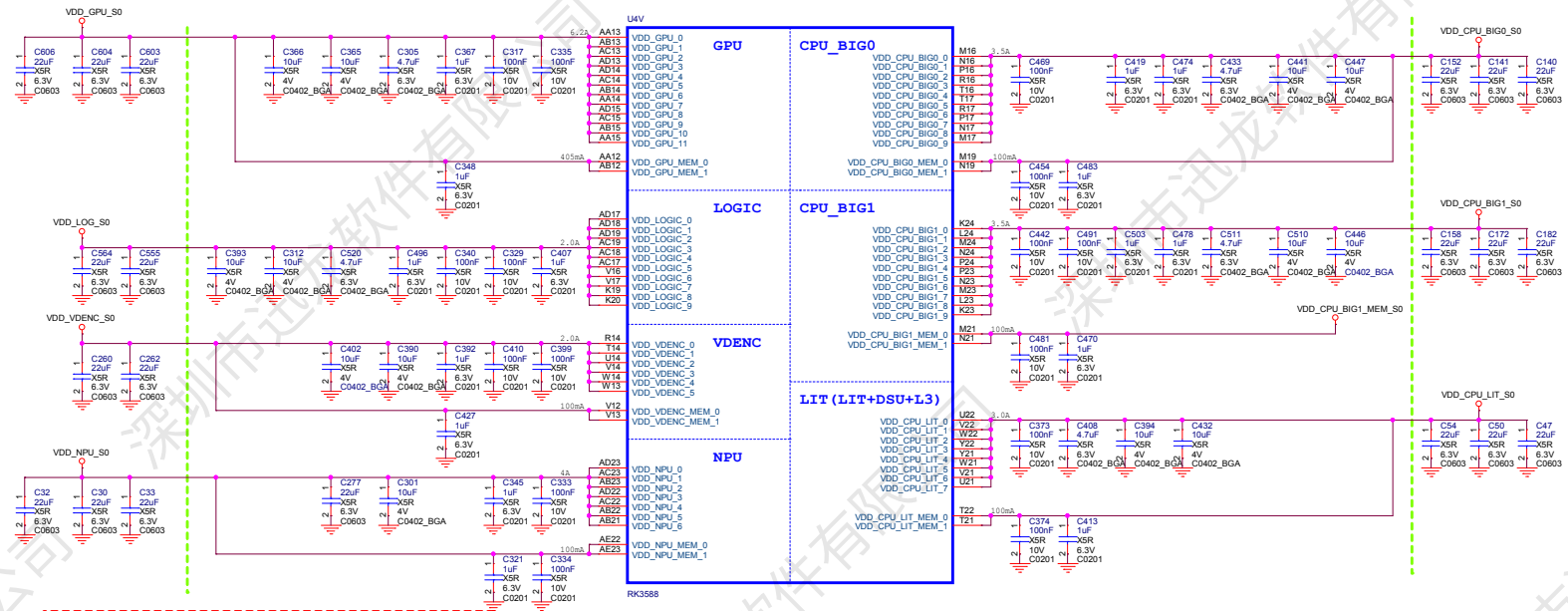
File: 08.RTC/FAN

Date:	Friday, July 25, 2025
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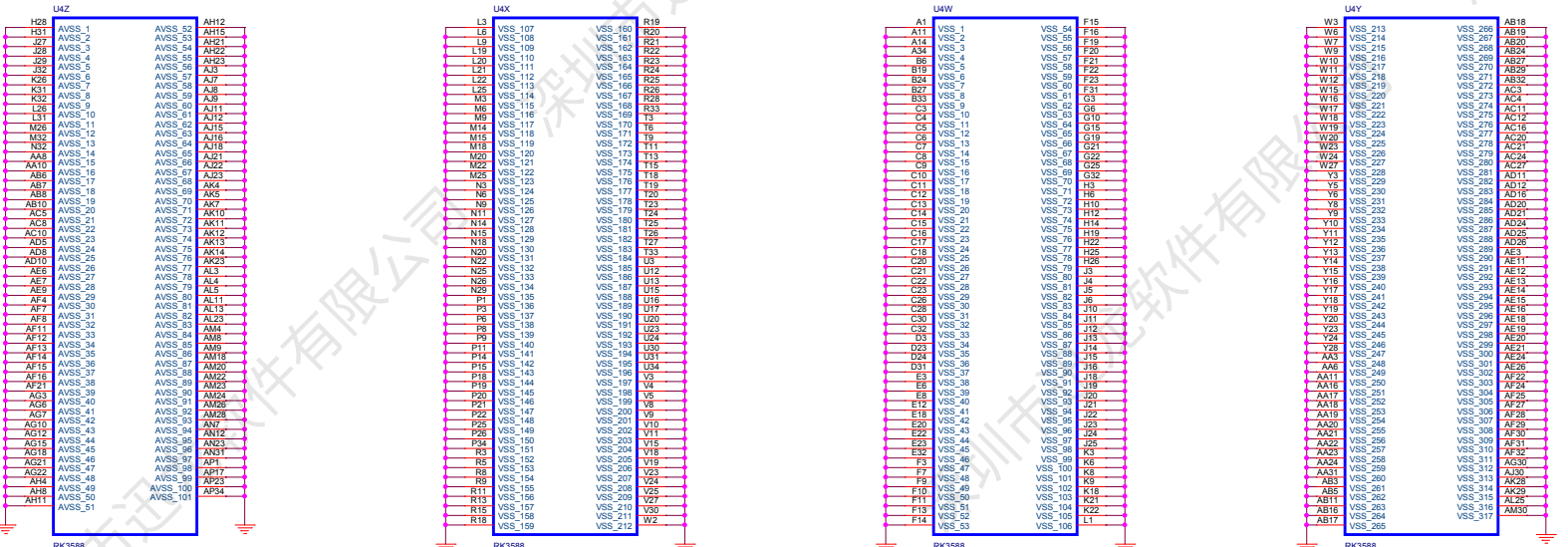
Rev:	V1.3.1
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Size:	A3	Sheet:	8	Of:	33
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Note:
The Caps between green line and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package



RK3588_E (OSC/PLL/PMUIO1/2)

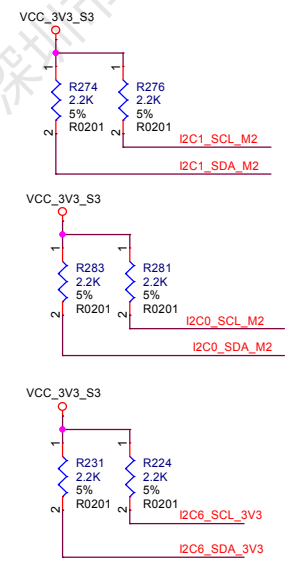
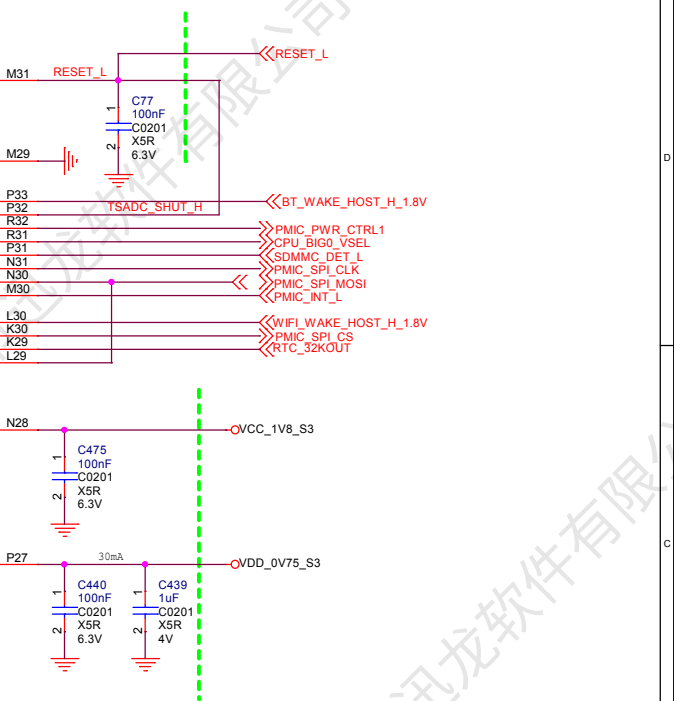
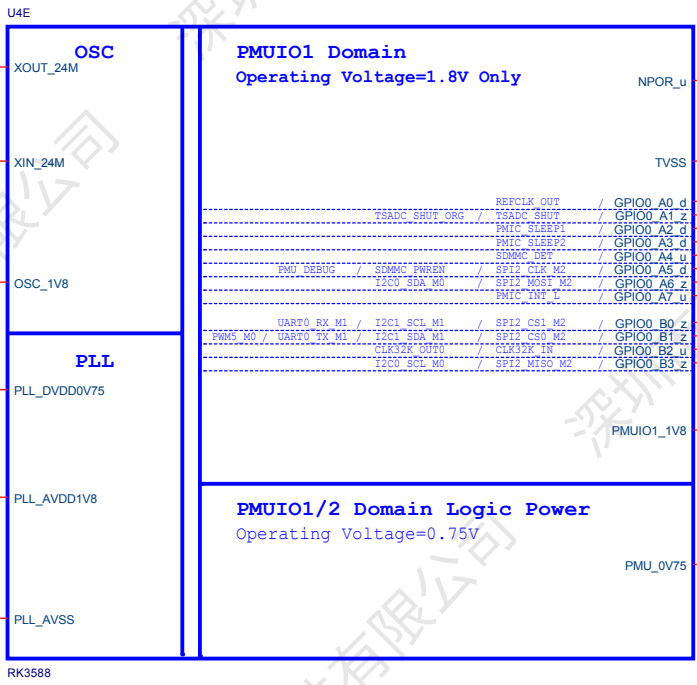
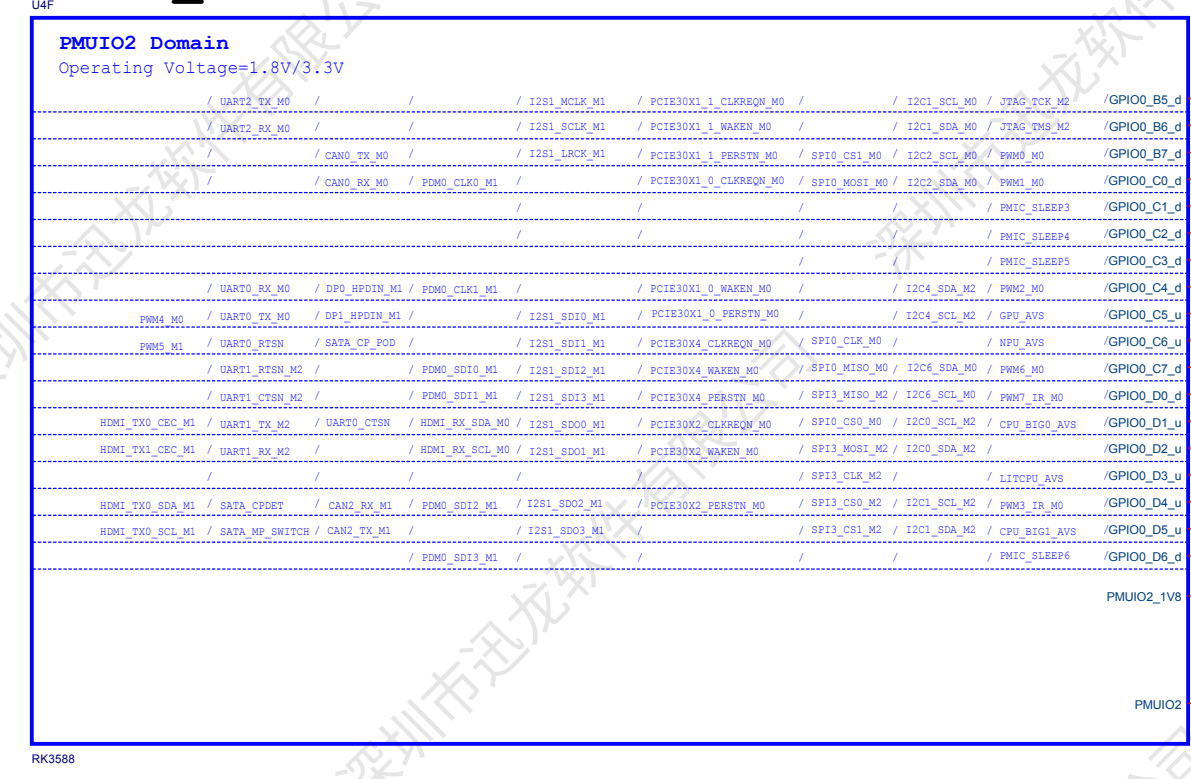
Note:
Adjusted the load capacitance according to the crystal specification

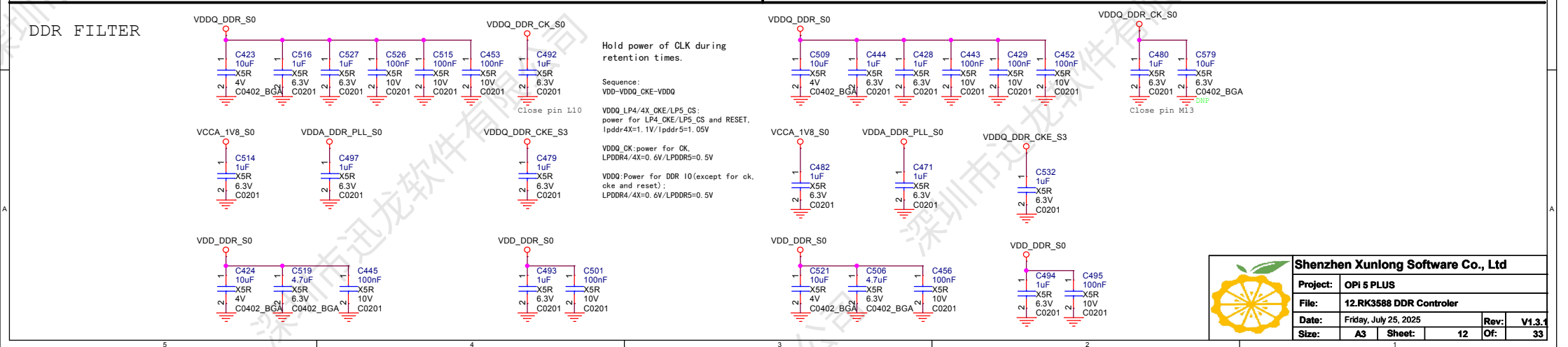
The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

$CL = \{CL1 * CL2 / (CL1 + CL2)\} + PCB \text{ strays}$
Total CL<=12pF

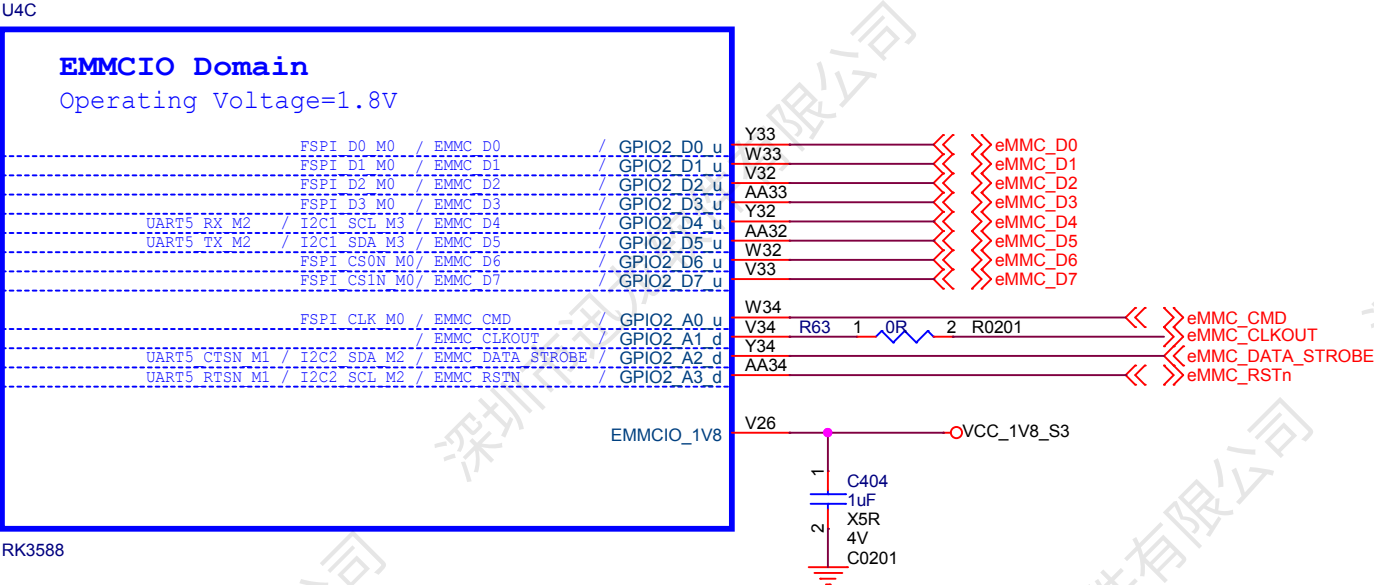
Note:
The Caps between green line and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

RK3588_F (PMUIO2)

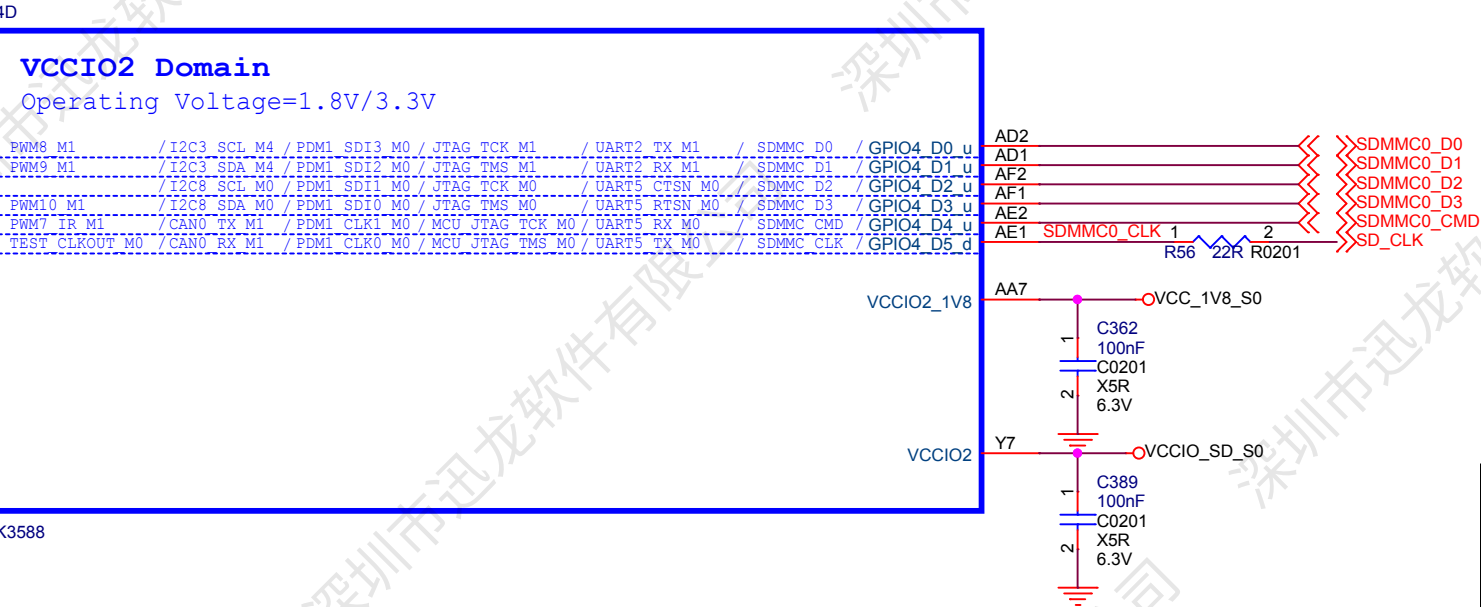




RK3588_C (EMMCIO Domain)



RK3588_D (VCCIO2 Domain)





Shenzhen Xunlong Software Co., Ltd			
Project:	OPI 5 PLUS		
File:	13.RK3588_Flash/SD Controller		
Date:	Friday, July 25, 2025	Rev:	V1.3
Size:	A3	Sheet:	13 Of: 33

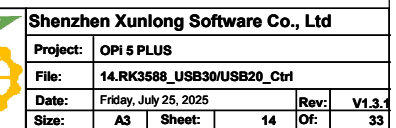
U4M



U4L



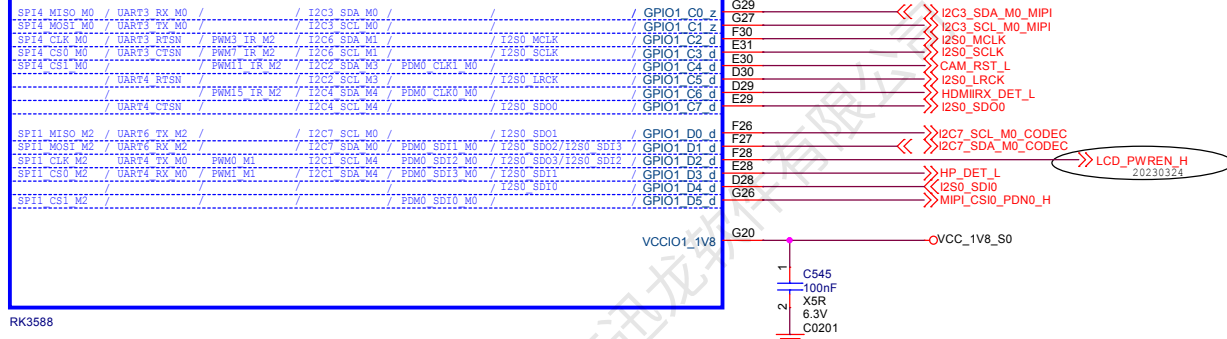
Option1	DP x4Lane	DP_TX Lane0-3
Option2	USB30 x4Lane	TYPEC_SSTX1P/1N&TYPEC_SSRX1P/1N or TYPEC_SSTX2P/2N&TYPEC_SSRX2P/2N
Option3	USB30X2Lane+DPX2Lane	USB30: Lane0 Lane1 DP: Lane2 Lane3
Option4	USB30X2Lane+DPX2Lane	USB30: Lane2 Lane3 DP: Lane0 Lane1



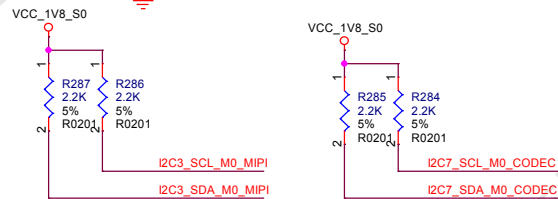
RK3588_G (VCCIO1 Domain)

U4G

VCCIO1 Domain
Operating Voltage=1.8V



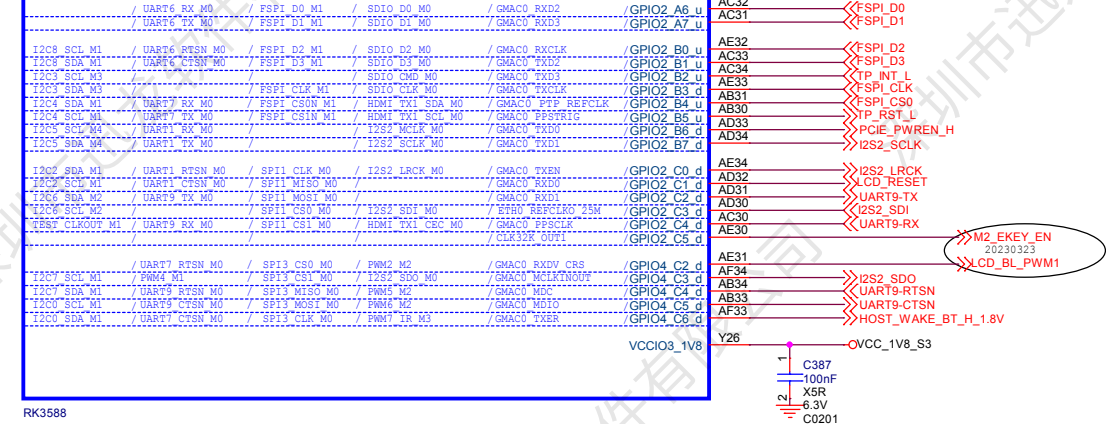
RK3588



RK3588_H (VCCIO3 Domain)

U4H

VCCIO3 Domain
Operating Voltage=1.8V



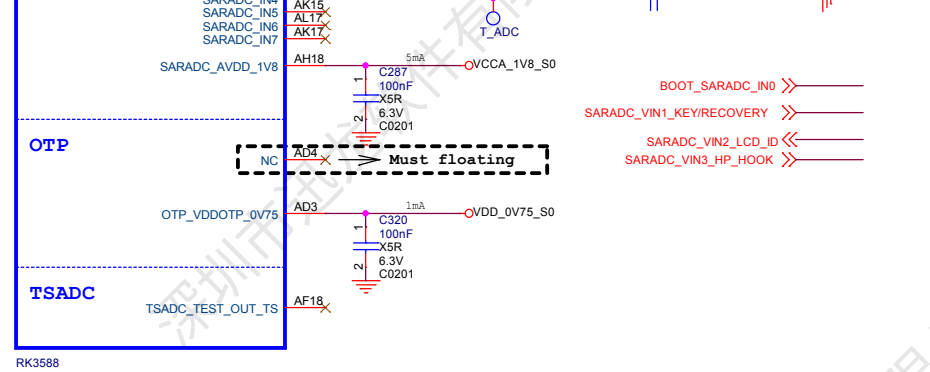
RK3588



RK3588_U (SARADC/OTP)

U4U

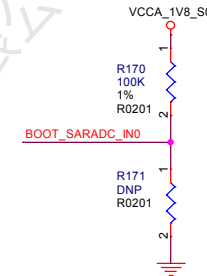
SARADC
12-bit 1MS/s



BOOT MODE CONFIG

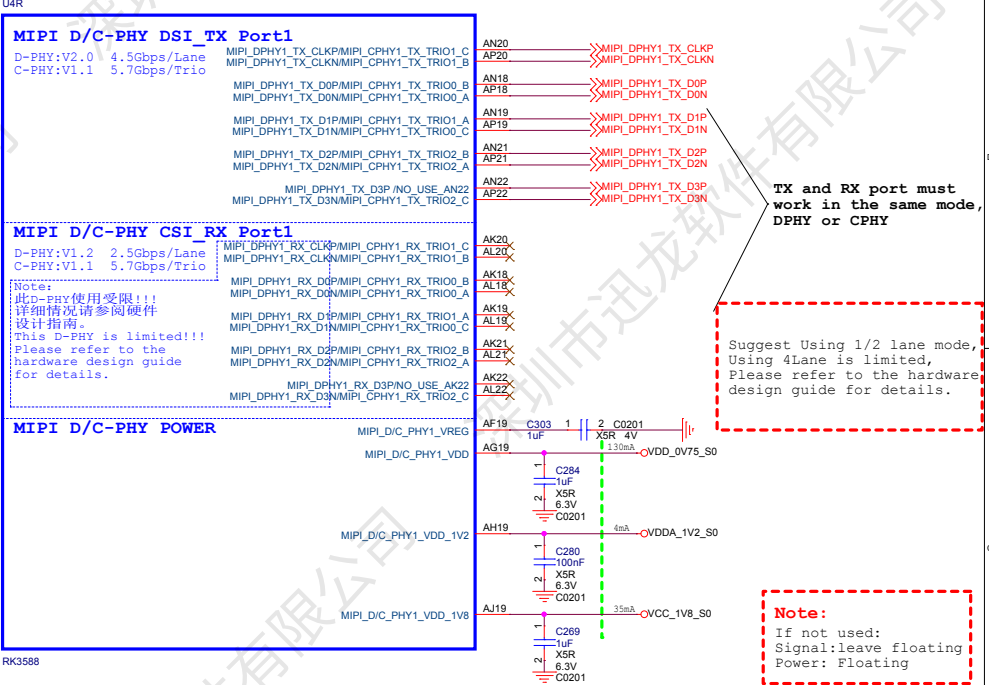
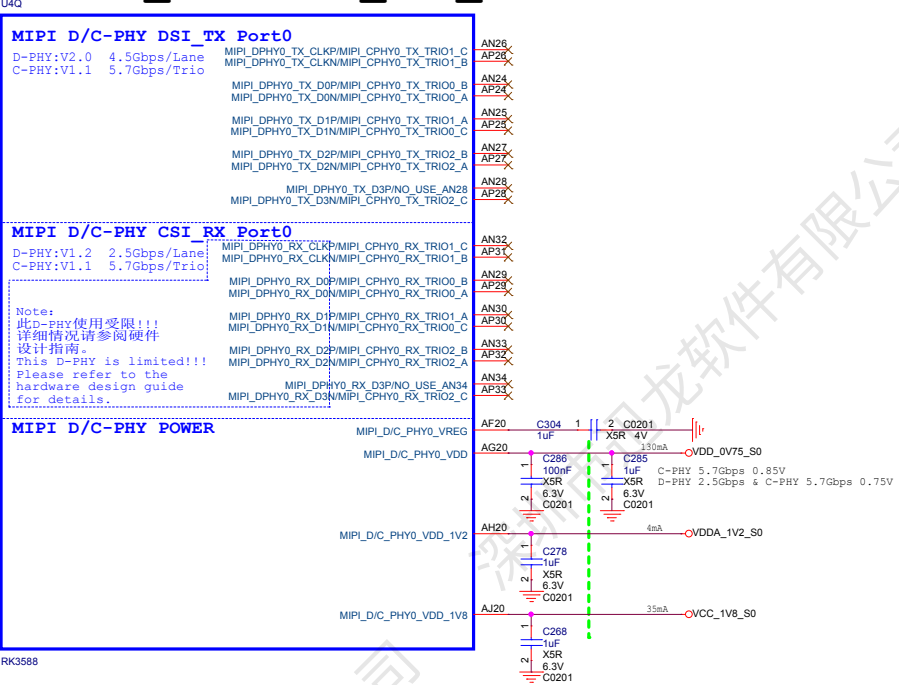
TABLE 1

Item	Rup	Rdown	ADC	VOL	BOOT MODE
LEVEL1	DNP	100K	0	0V	USB (Maskrom mode)
LEVEL2	100K	20K	682	0.3V	SD Card-USB
LEVEL3	100K	51K	1365	0.6V	EMMC-USB
LEVEL4	100K	100K	2047	0.9V	FSPI M0-USB
LEVEL5	100K	200K	2730	1.2V	FSPI M1-USB
LEVEL6	100K	499K	3412	1.5V	FSPI M2-USB
LEVEL7	100K	DNP	4095	1.8V	FSPI M2-FSPI M1-FSPI M0-EMMC-SD Card-USB

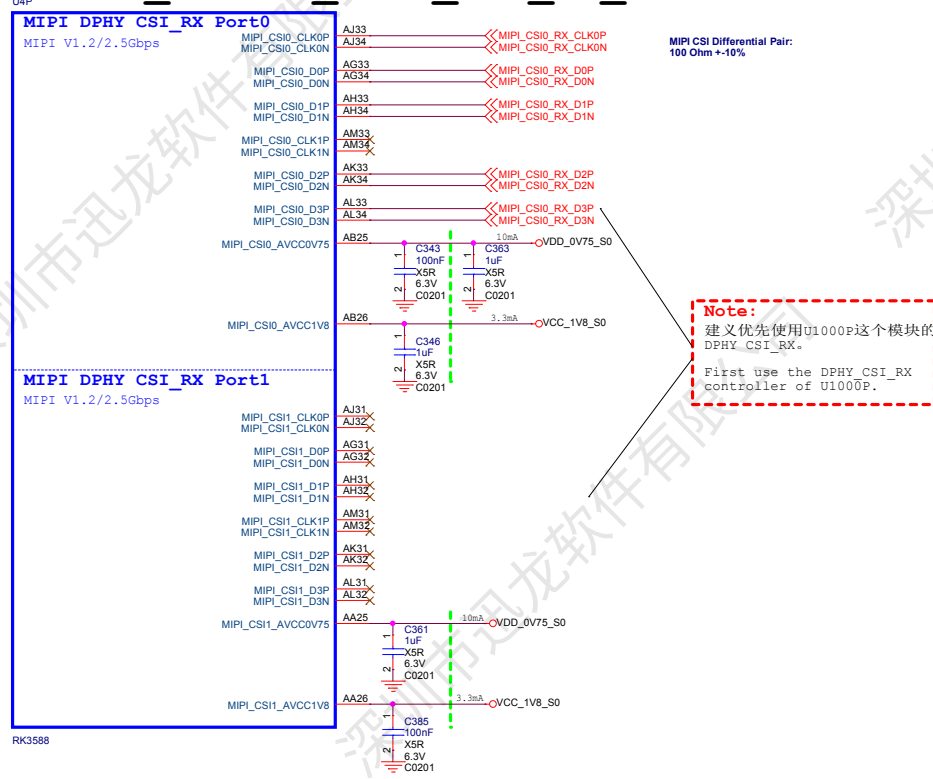


Shenzhen Xunlong Software Co., Ltd			
Project:	OPI 5 PLUS		
File:	15.RK3588_SARADC/1.8V Only GPIO		
Date:	Friday, July 25, 2025	Rev:	V1.3.1
Size:	A3	Sheet:	15
Of:			33

RK3588_Q/R(MIPI_D/C_PHY0/1)



RK3588_P(MIPI_DPHY_CSI_RX_PHY)



MIPI_CSI_RX Configuration

Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

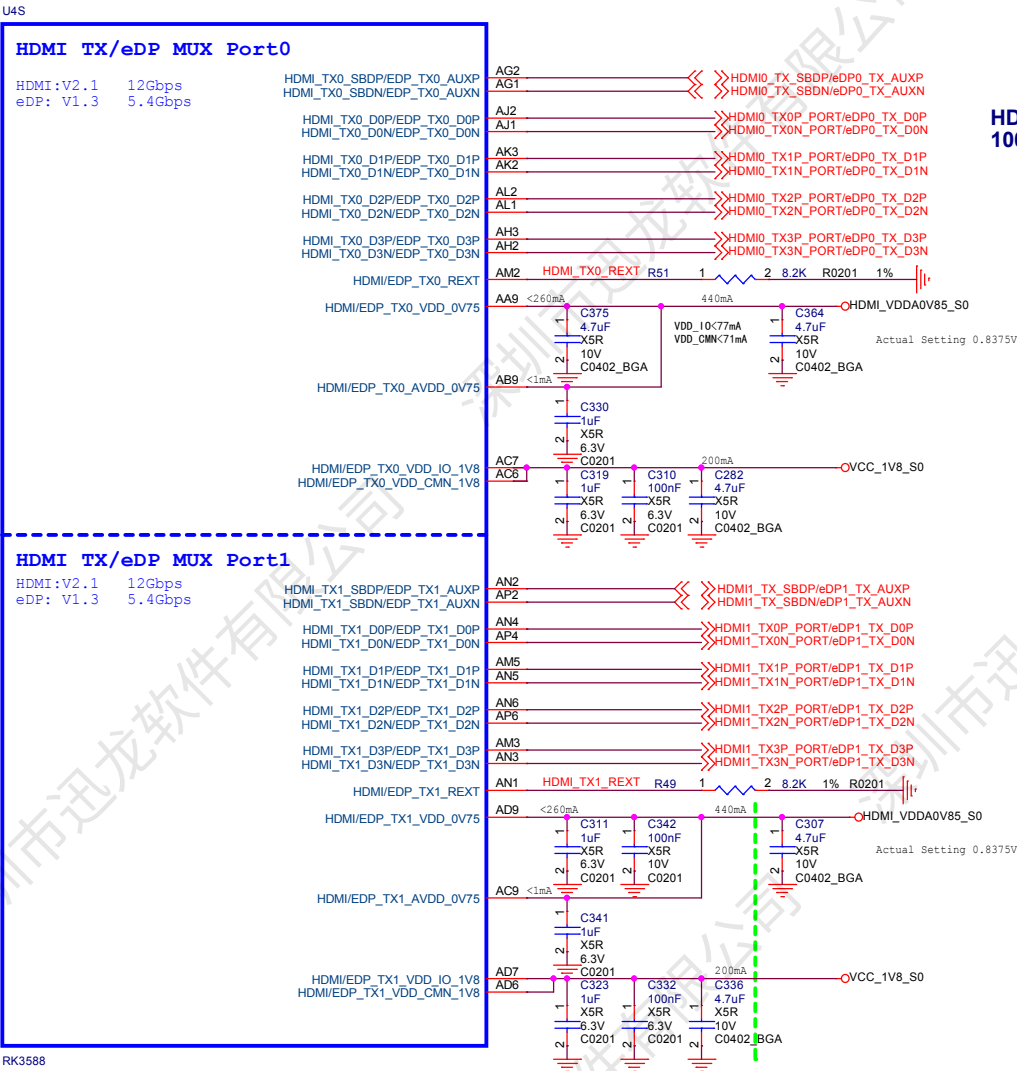
Note:
When in single clock lane mode, CLK0P/0N is the clock lane from Data lane0 to Data lane3, but clock lane1 is invalid; In dual clock lanes mode, CLK0P/0N is the clock lane of Data lane0 and Data lane1, while CLK1P/1N is the clock lane of Data lane2 and Data lane3.

Note:
The Caps to the left of green line should be placed under the U1000 package. Other caps should be placed close to the U1000 package.

Note:
If not used:
Signal:leave floating
Power: Floating

RK3588_S (HDMI2.1 TX)

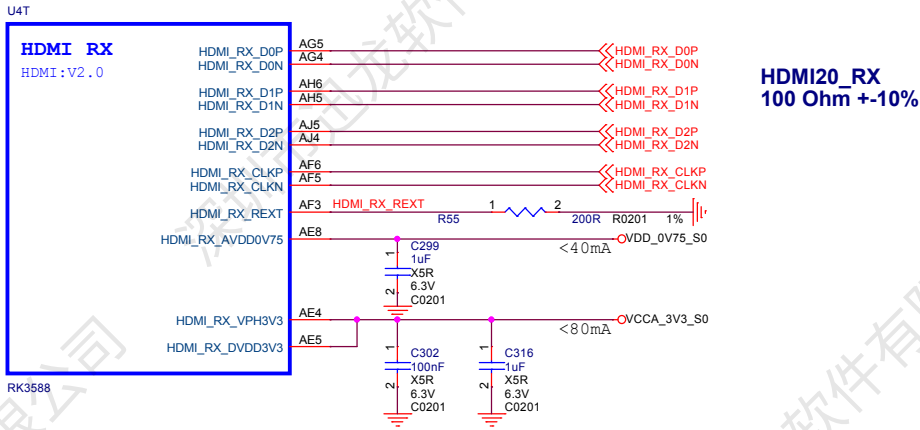
Note:
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.



Note:
The Caps to the left of green line should be placed under the U1000 package. Other caps should be placed close to the U1000 package.

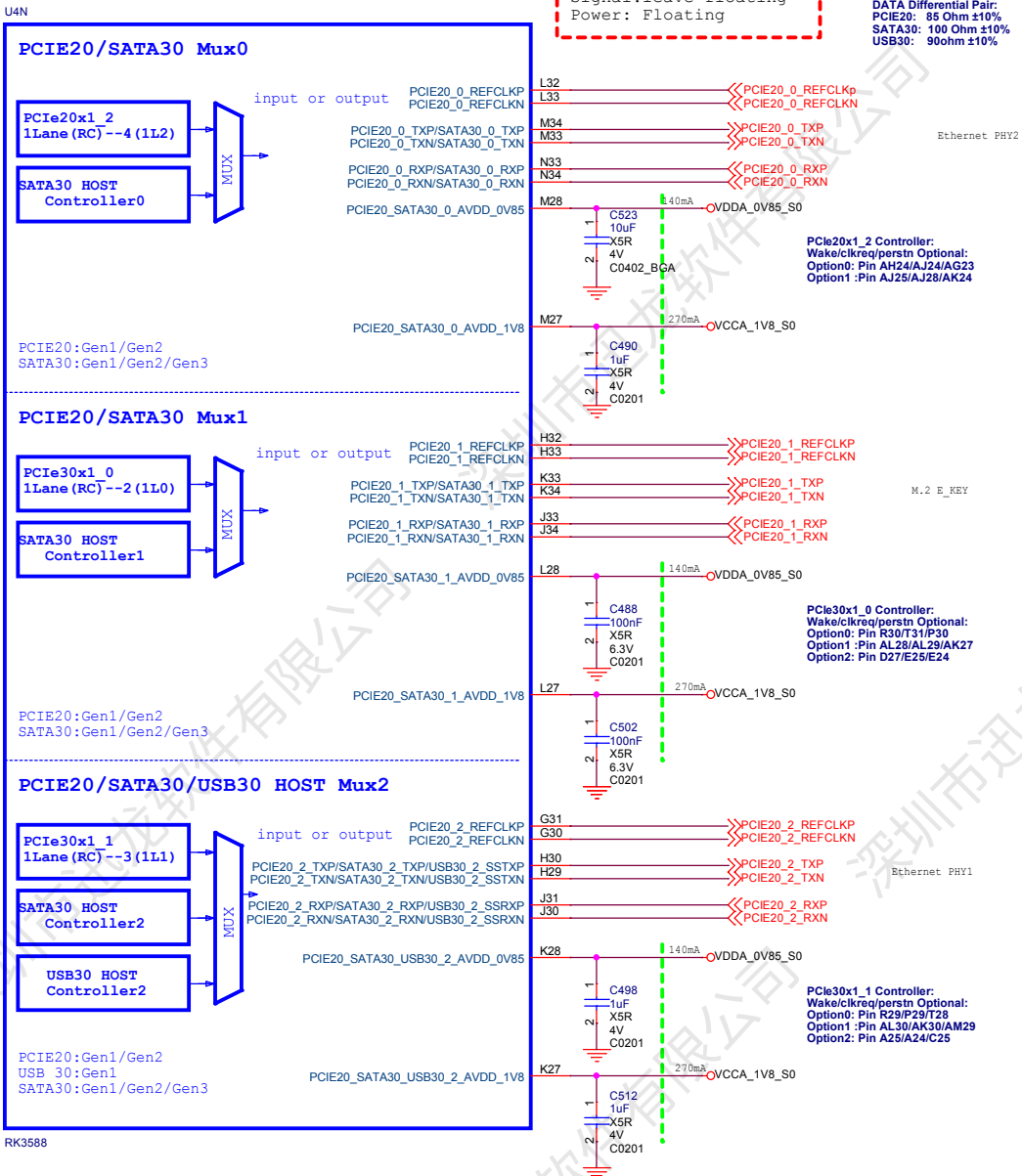
Note:
If not used:
Signal: leave floating
Power: Floating or tie to VSS

RK3588_T (HDMI20 RX)



Note:
If not used:
Signal: leave floating
Power: Floating

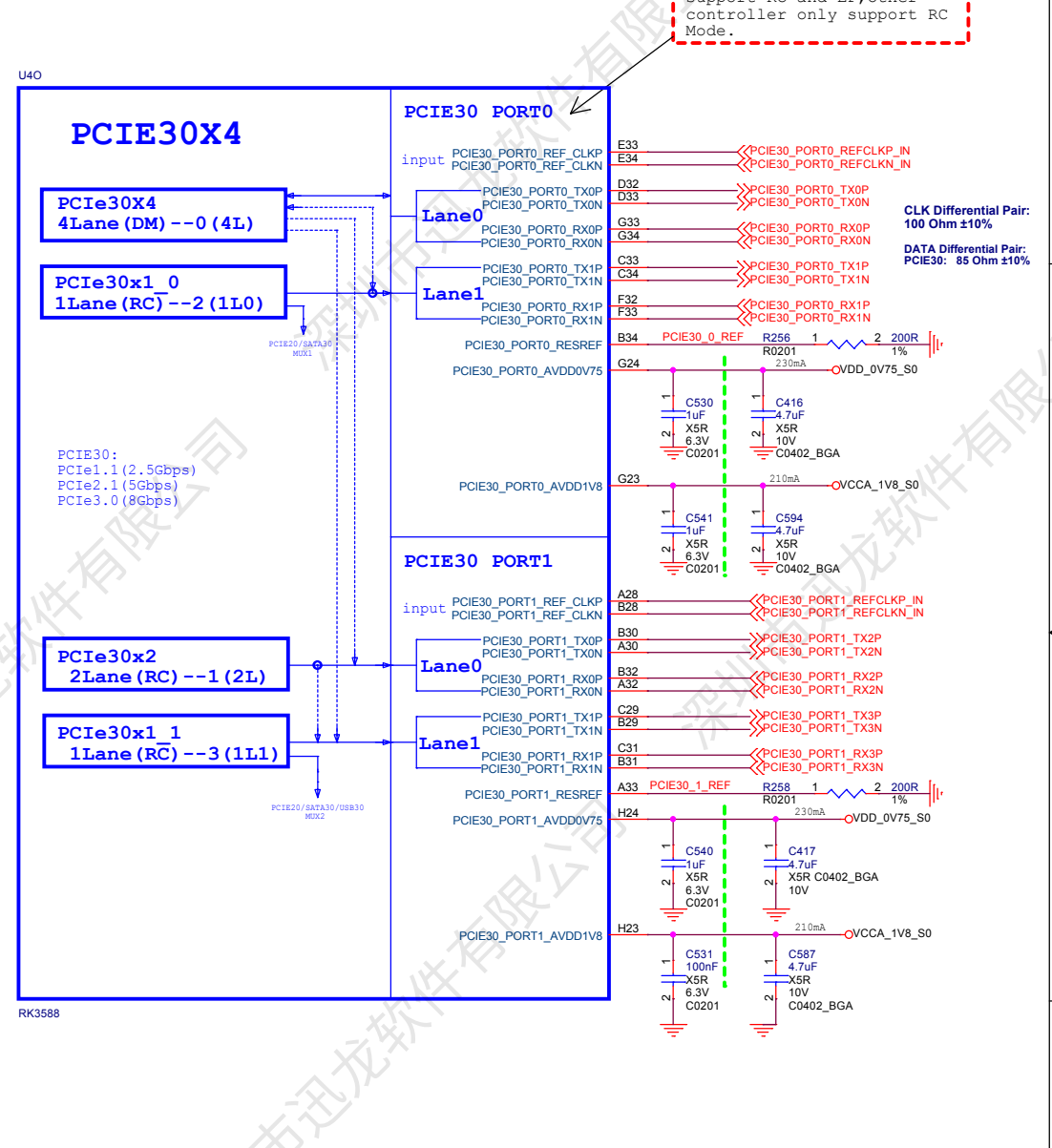
RK3588_N (PCIE20)



Note:
The SATA differential trace impedance is 100 OHM
The SATA trace length is less than 5 inch

Note:
每个SATA PORT最多只能扩展5个SATA，不支持二级扩展。
Each sata port can only expand 5 sata port at most, No support level-2 expand.

RK3588_O (PCIE30)



Note:
Only PCIE3.0 Controller 0 support RC and EP, Other controller only support RC Mode.

Note:
CLK Differential Pair:
100 Ohm ±10%

Note:
DATA Differential Pair:
PCIE30: 85 Ohm ±10%

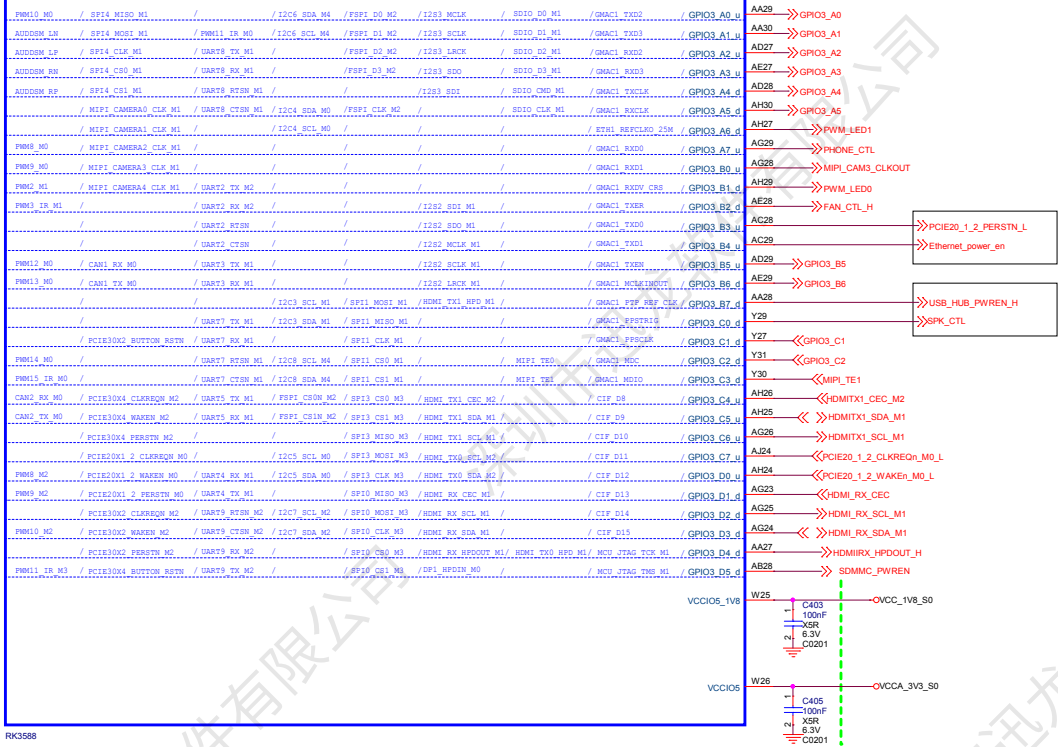
				Shenzhen Xunlong Software Co., Ltd			
Project:		OPI 5 PLUS		File:		18.RK3588_PCIE30/PCIE20/SATA30	
Date:		Friday, July 25, 2025		Rev:		V1.3.1	
Size:		A3		Sheet:		18	
				Of:		33	

RK3588_J (VCCIO5 Domain)

U4J

VCCIO5 Domain

Operating Voltage=1.8V/3.3V



RK3588_K (VCCIO6 Domain)

U4K

VCCIO6 Domain

Operating Voltage=1.8V/3.3V

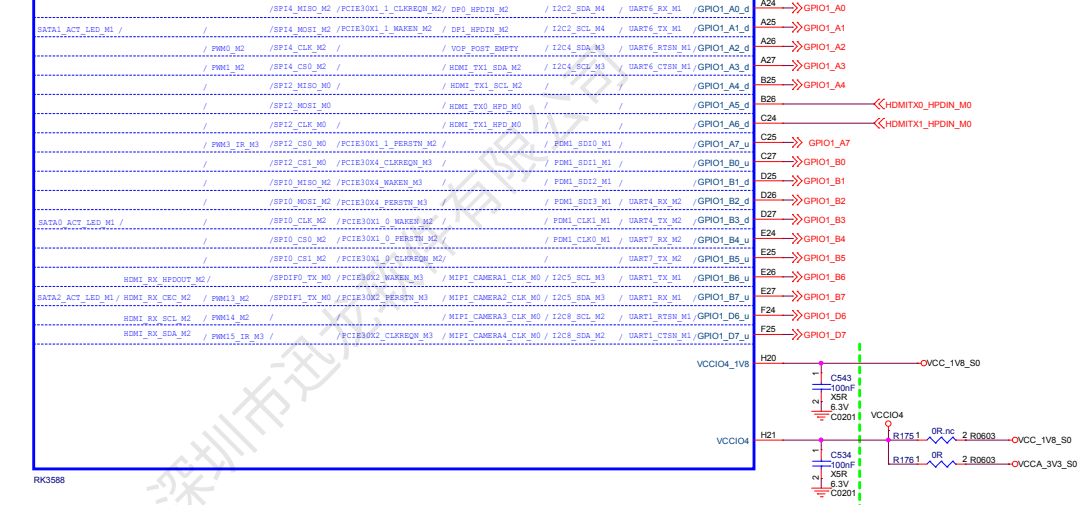


RK3588_I (VCCIO4 Domain)

U4I

VCCIO4 Domain

Operating Voltage=1.8V/3.3V



D

C

B

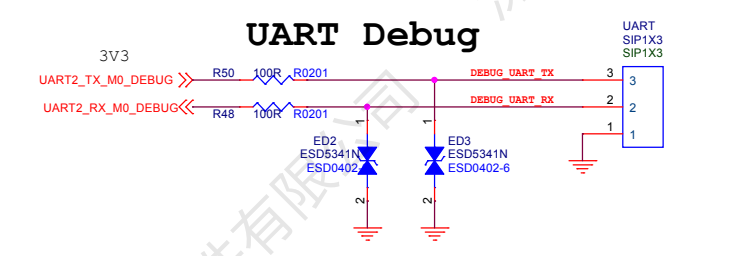
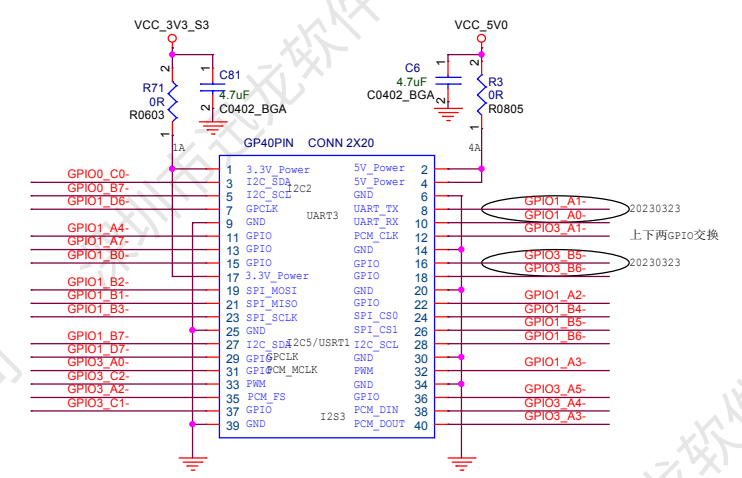
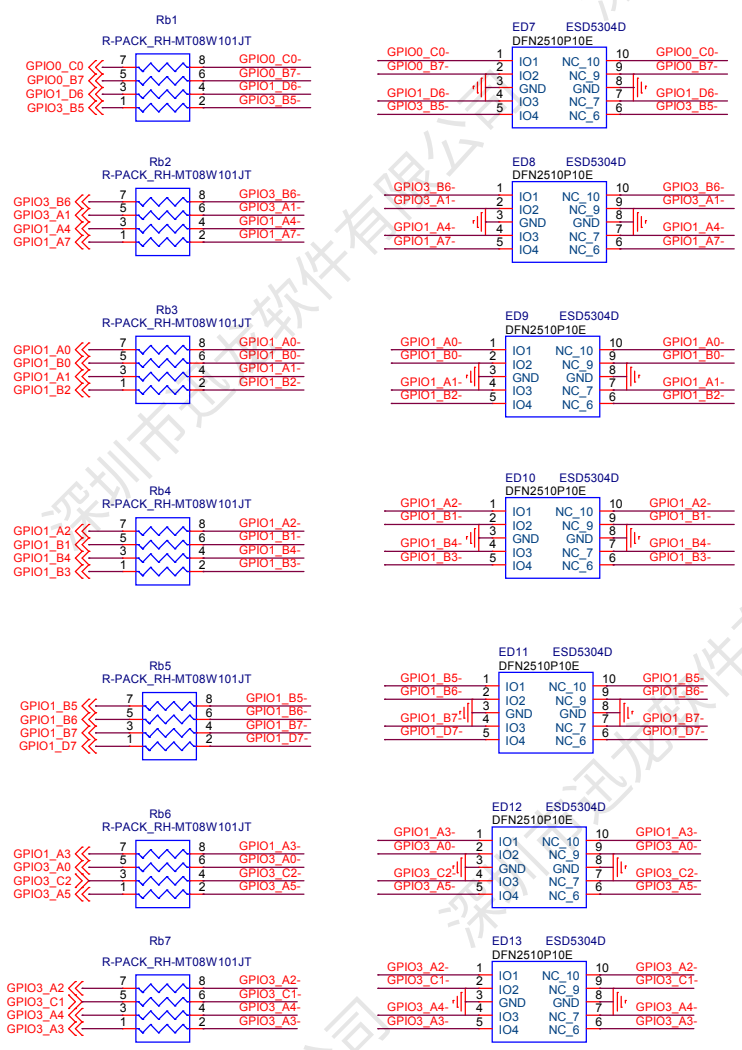
A

D

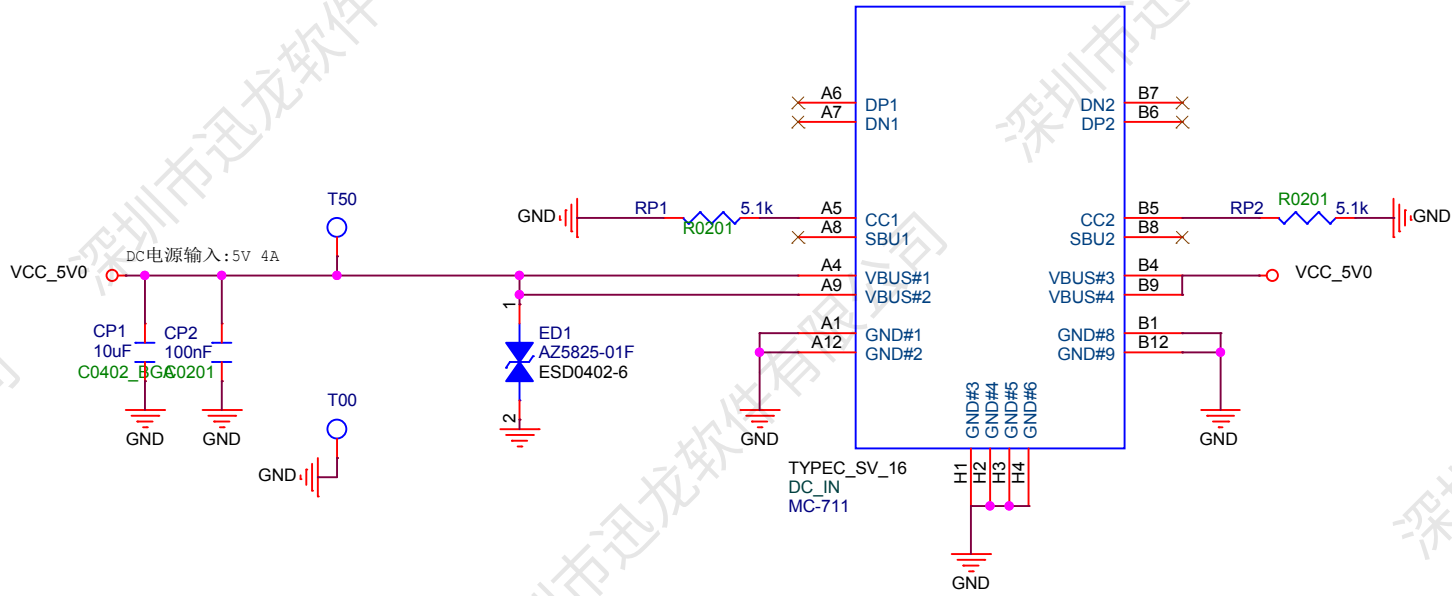
C

B

A

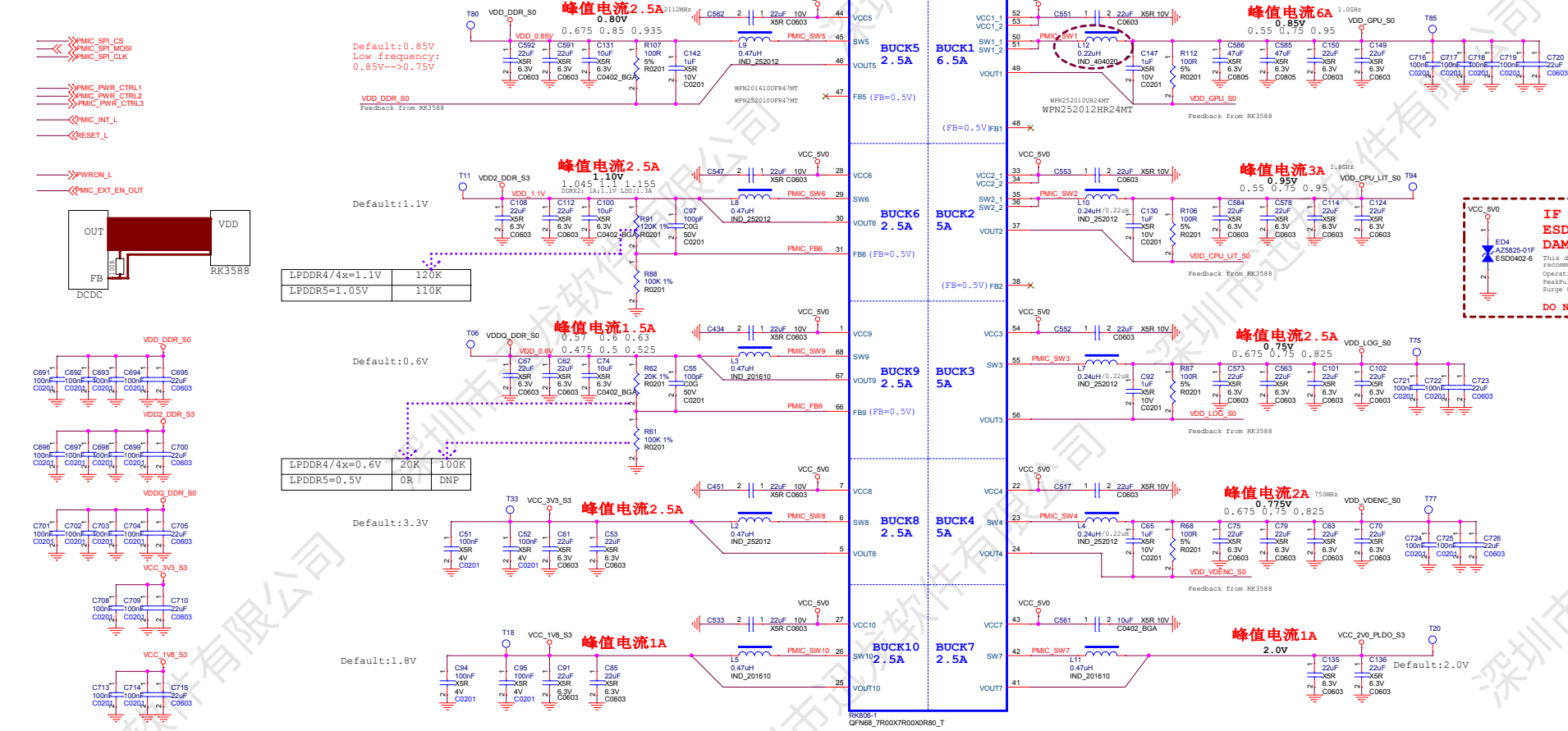


TPYE-C POWER IN

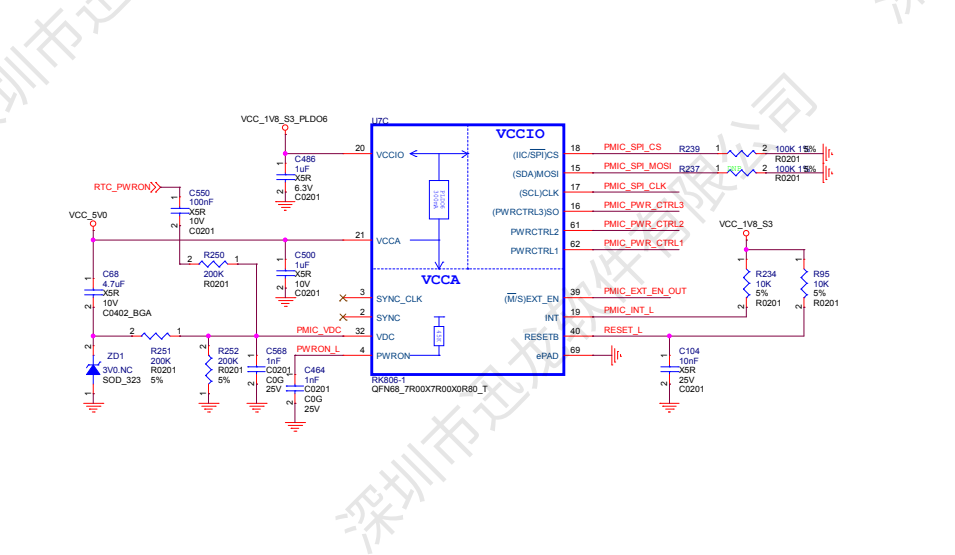


Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	21.Power_IN TYPE_C DC5V_4A			
Date:	Friday, July 25, 2025			Rev: V1.3.1
Size:	A3	Sheet:	21	Of: 33

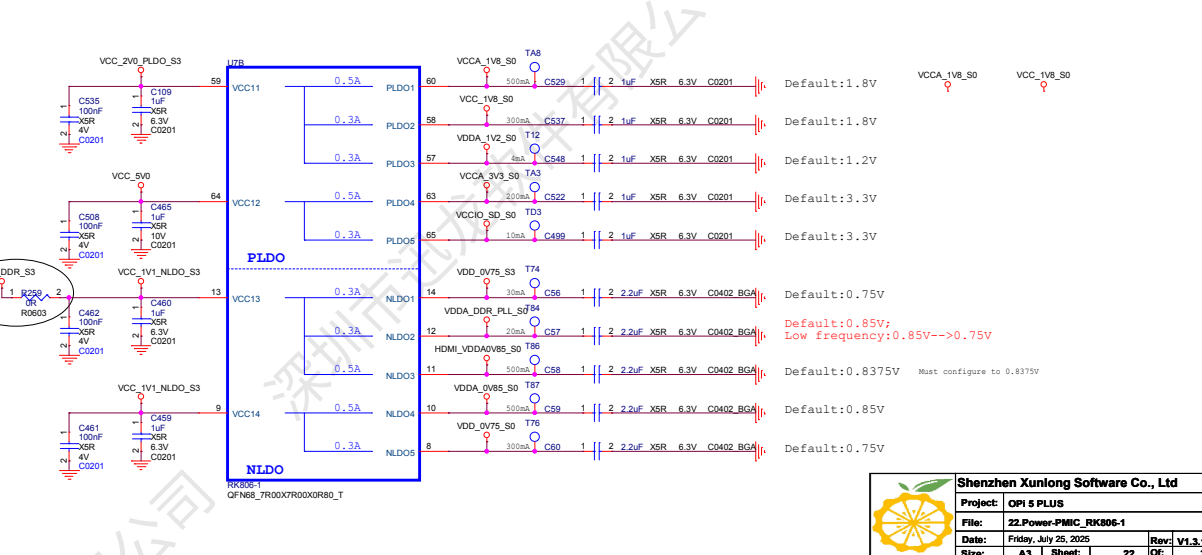
PMIC RK806-1 BUCK



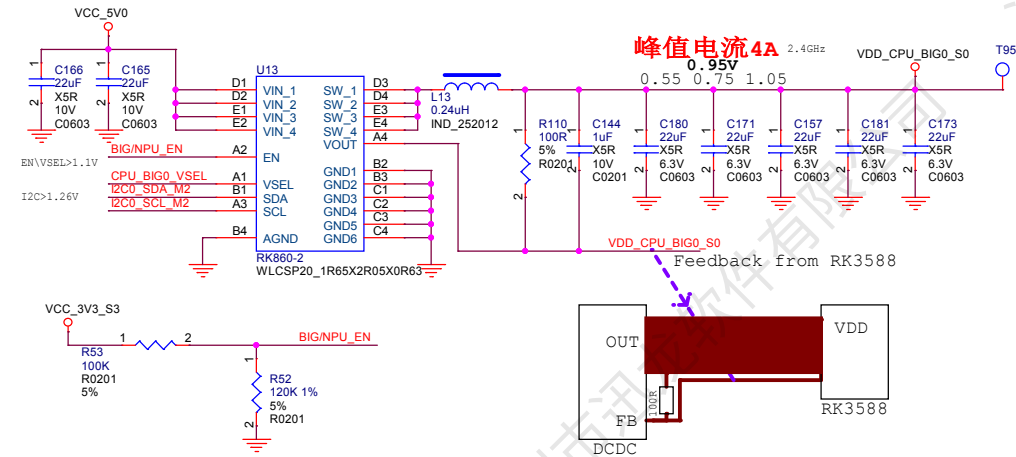
PMIC RK806-1 Managerment



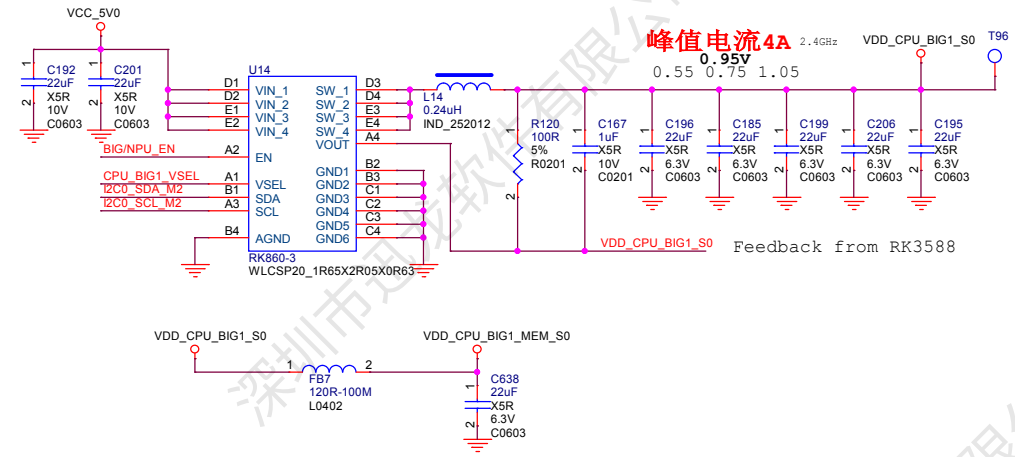
PMIC RK806-1 LDO



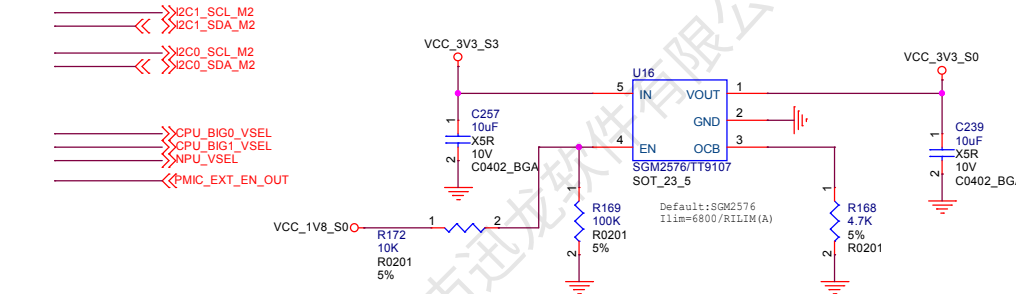
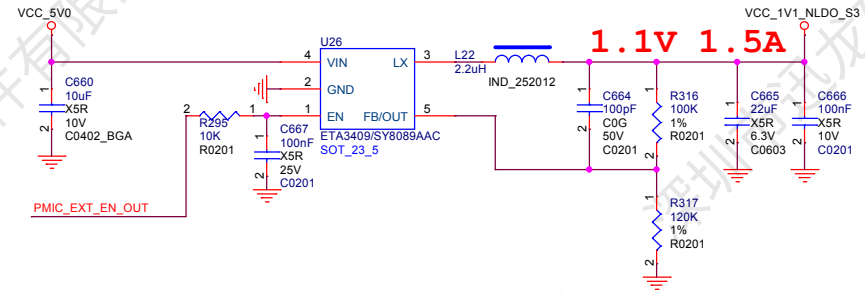
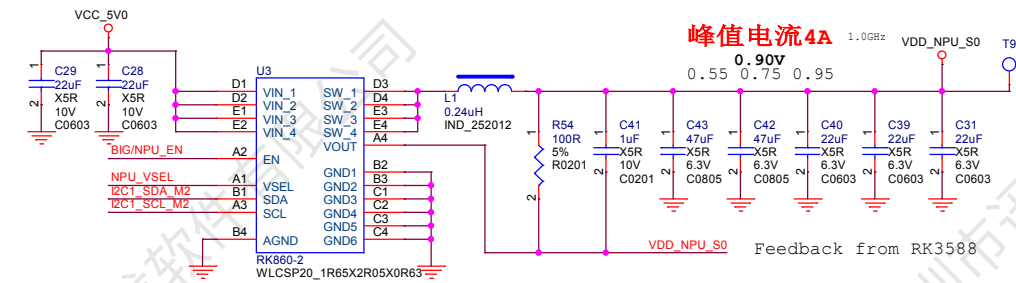
VDD_CPU_BIG0



VDD_CPU_BIG1

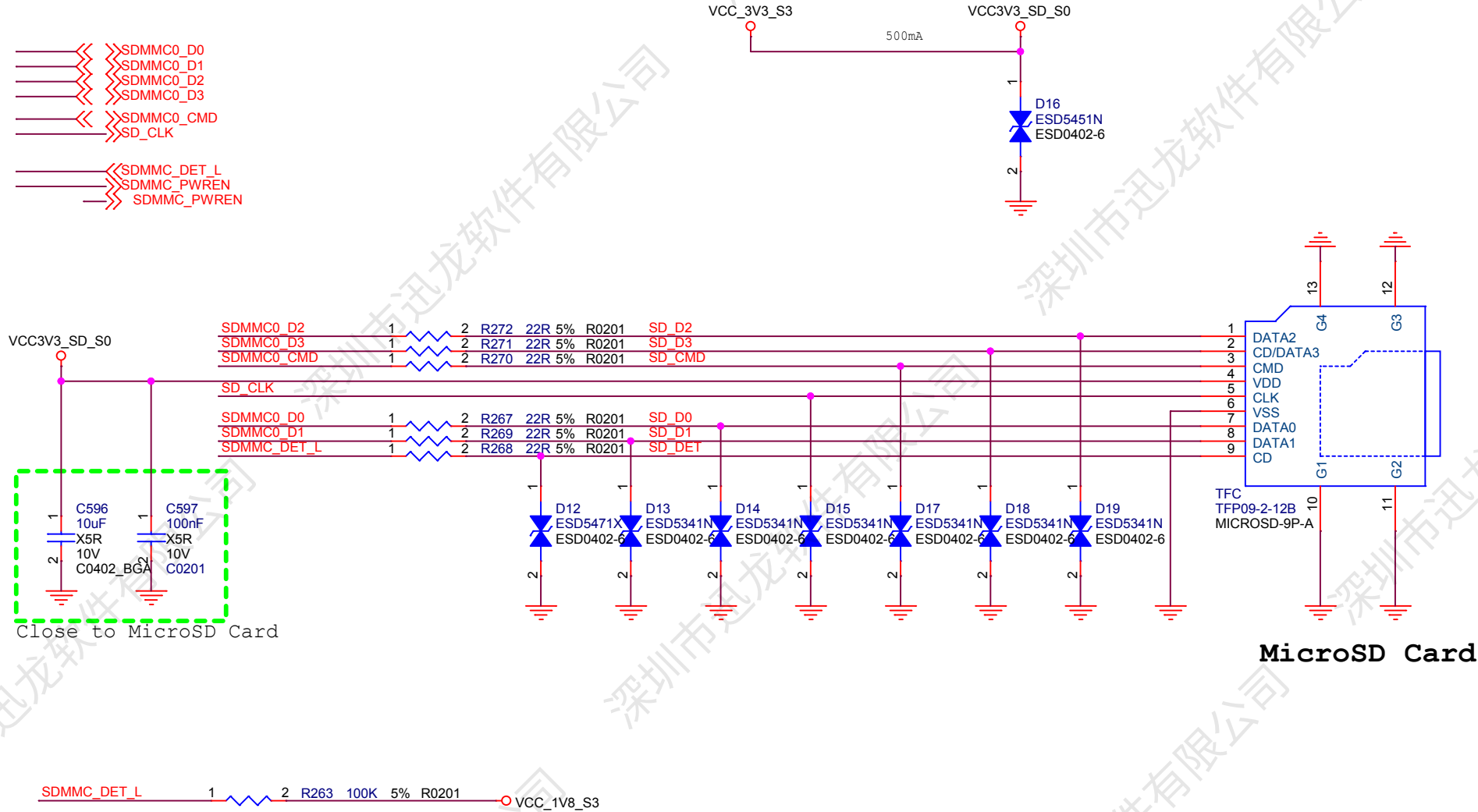


VDD_NPU



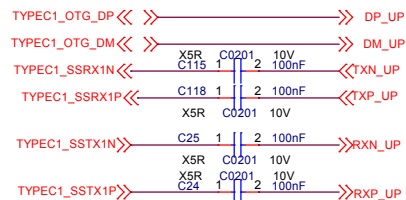
	Shenzhen Xunlong Software Co., Ltd				
	Project:	OPI 5 PLUS			
	File:	23.Power_Ext Discrete			
	Date:	Friday, July 25, 2025			Rev: V1.3.1
	Size:	A3	Sheet:	23	Of: 33

TF CARD

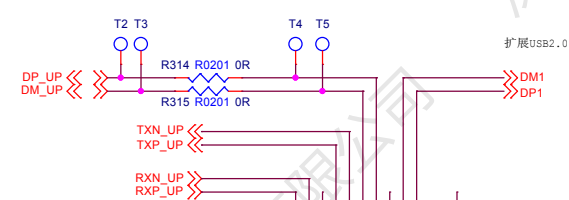


Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	24.Flash-TF Card			
Date:	Friday, July 25, 2025	Rev:	V1.3.1	
Size:	A3	Sheet:	24	Of: 33

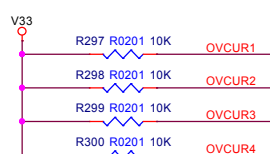
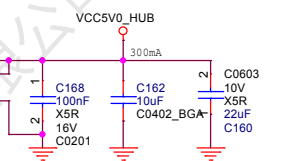
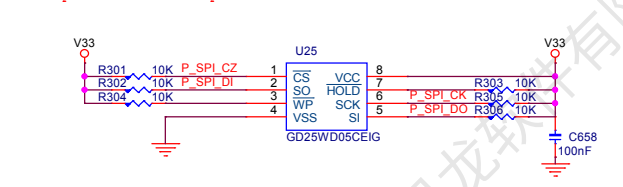
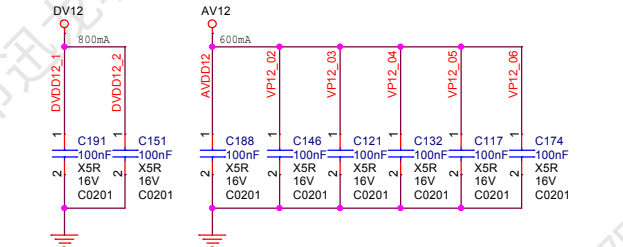
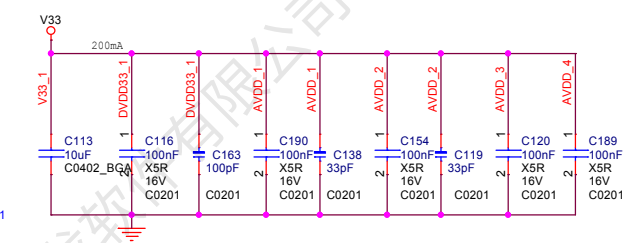
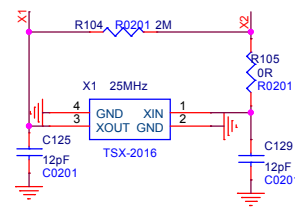
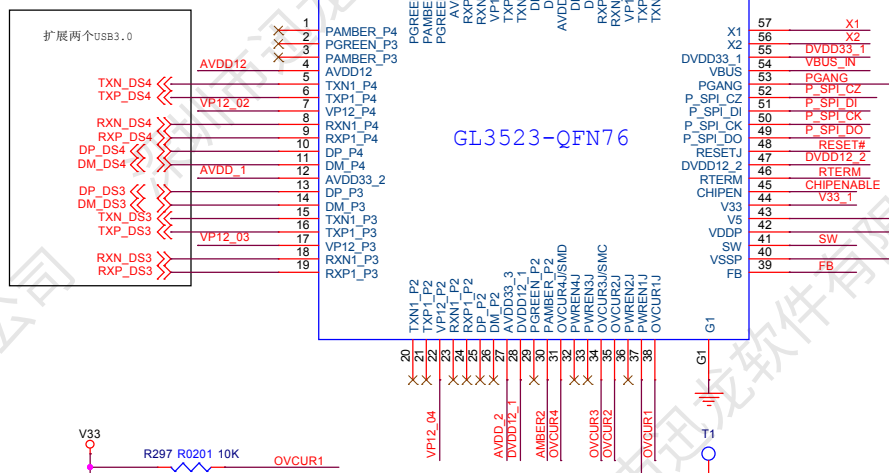
USB30 HOST PORT



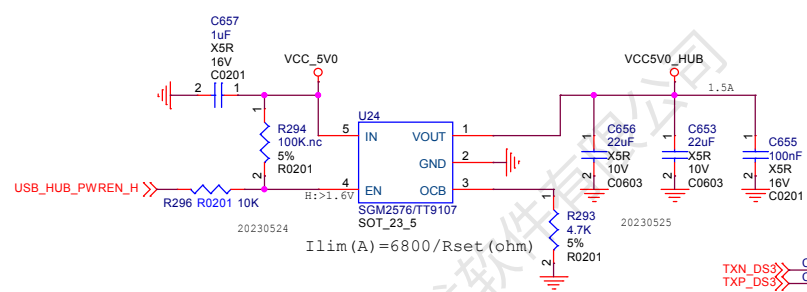
USB3.0 for HOST



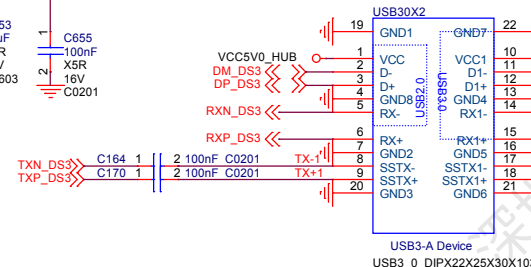
U11 USB3.0 Hub



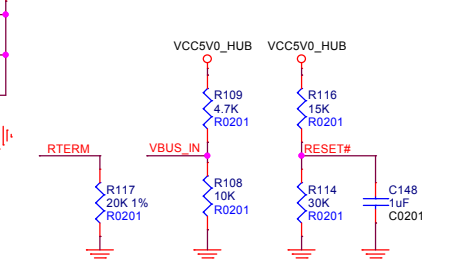
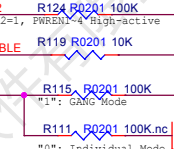
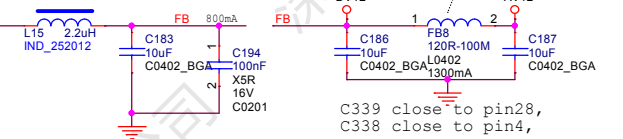
PWREN1# is the only power-enable output for GANG-mode.

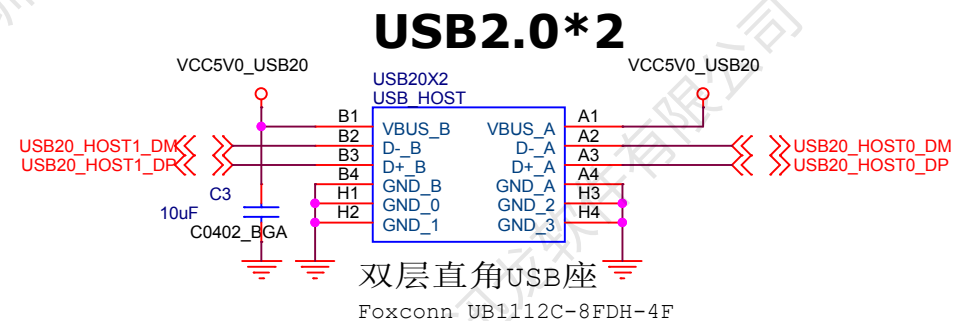
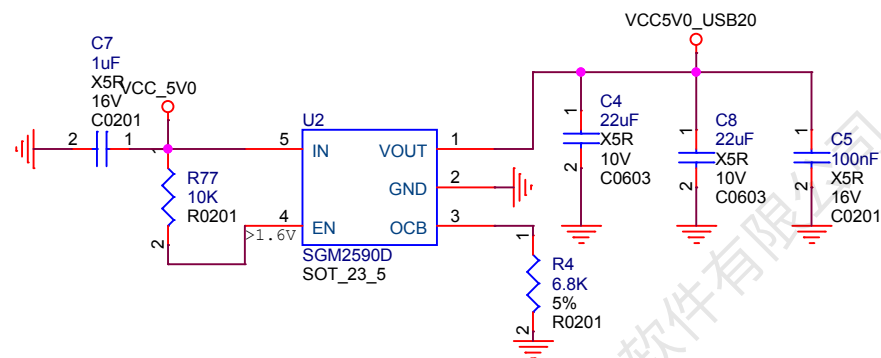


USB3.0x2插座

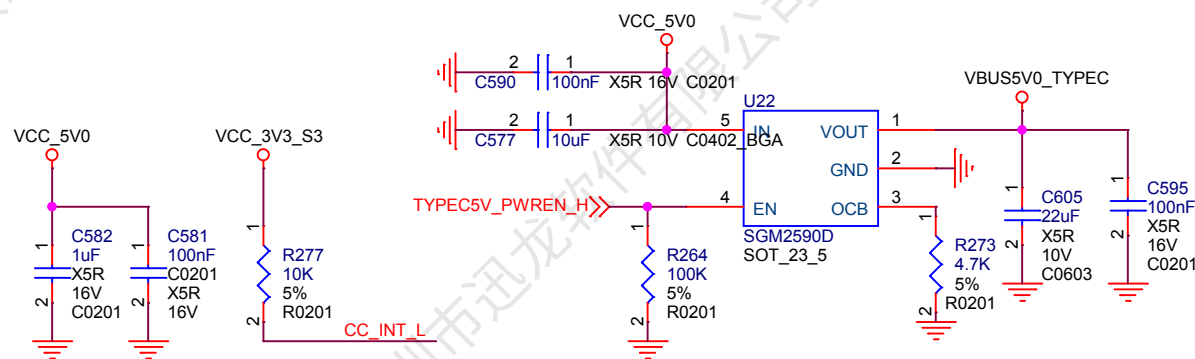
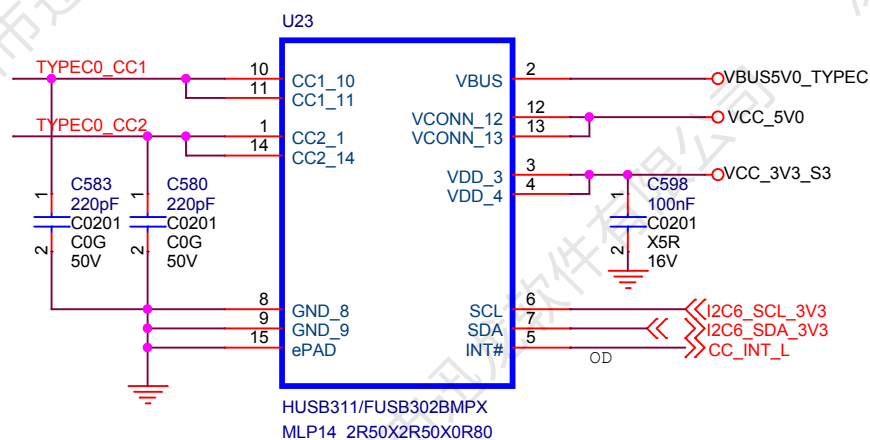
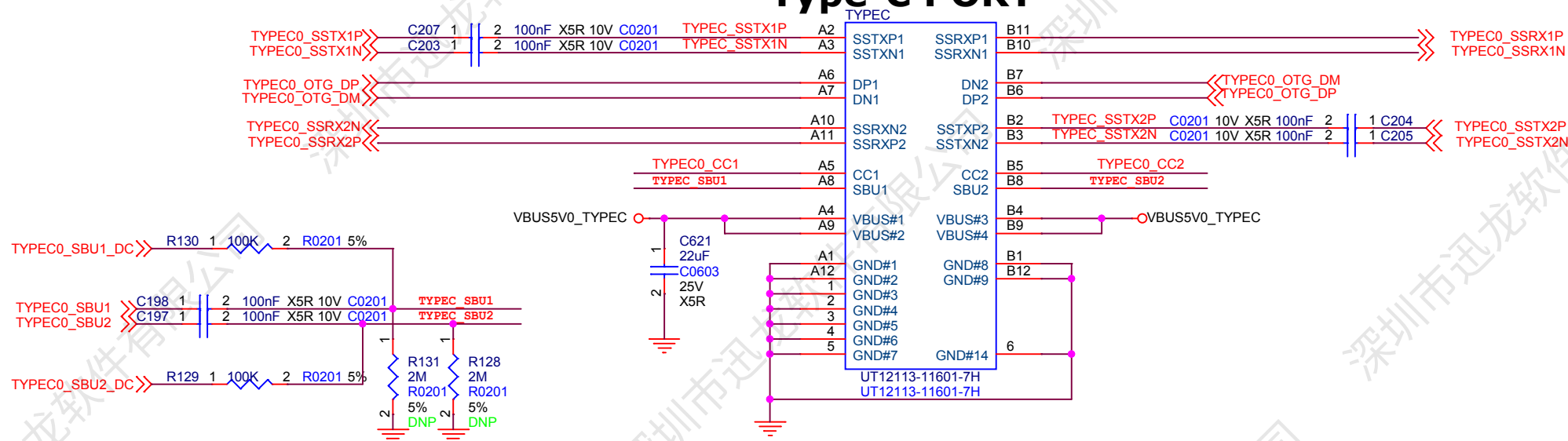


1.2v CIRCUIT





Type-C PORT



Shenzhen Xunlong Software Co., Ltd

Project:	OPI 5 PLUS
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File:	26.Type C/USB2.0
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Date:	Friday, July 25, 2025
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Size:	A3	Sheet:	26	Of:	33
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Of:	3
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3:

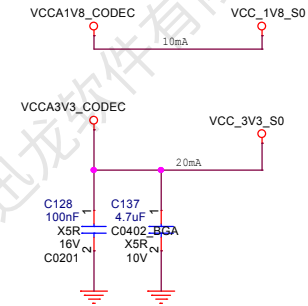
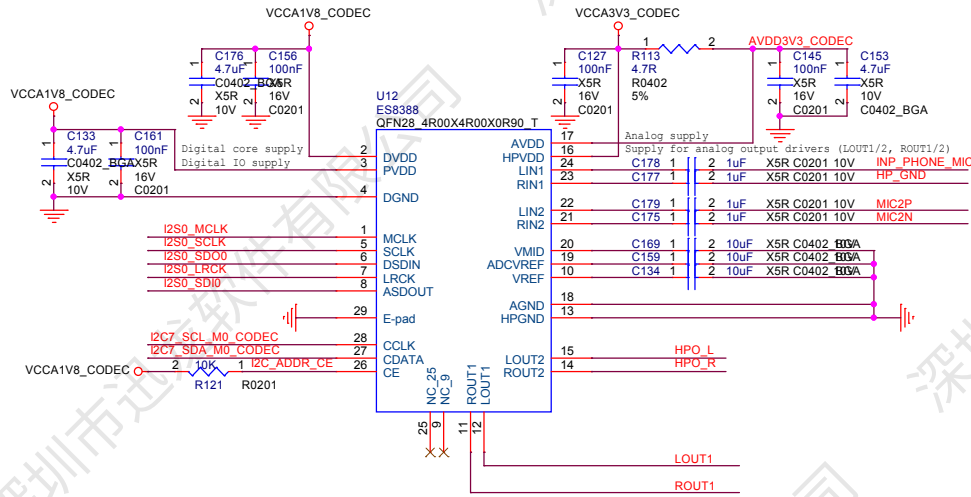
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CODEC ES8388

I2C7_SDA_M0_CODEC
I2C7_SCL_M0_CODEC
I2S0_MCLK
I2S0_SCLK
I2S0_LRCK
I2S0_SDO0
I2S0_SDIO

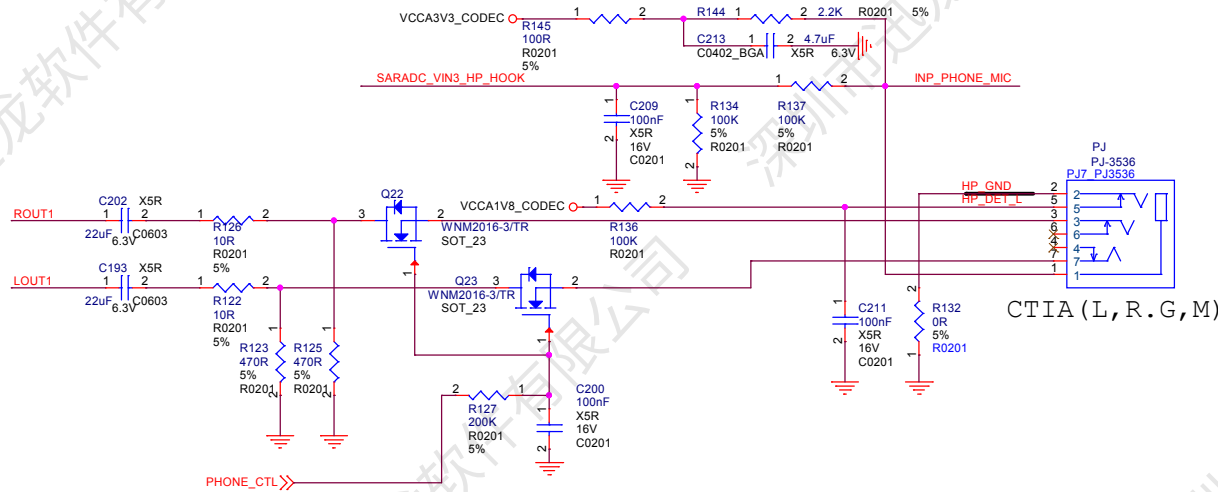
HPO_R
HPO_L

SARADC_VIN3_HP_HOOK
HP_DET_L
PHONE_CTL



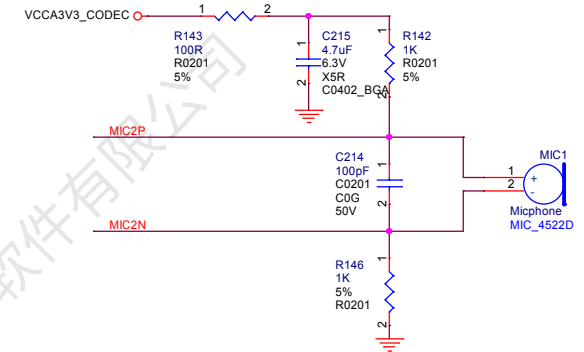
EARPHONE

CTIA (L, R, G, M)



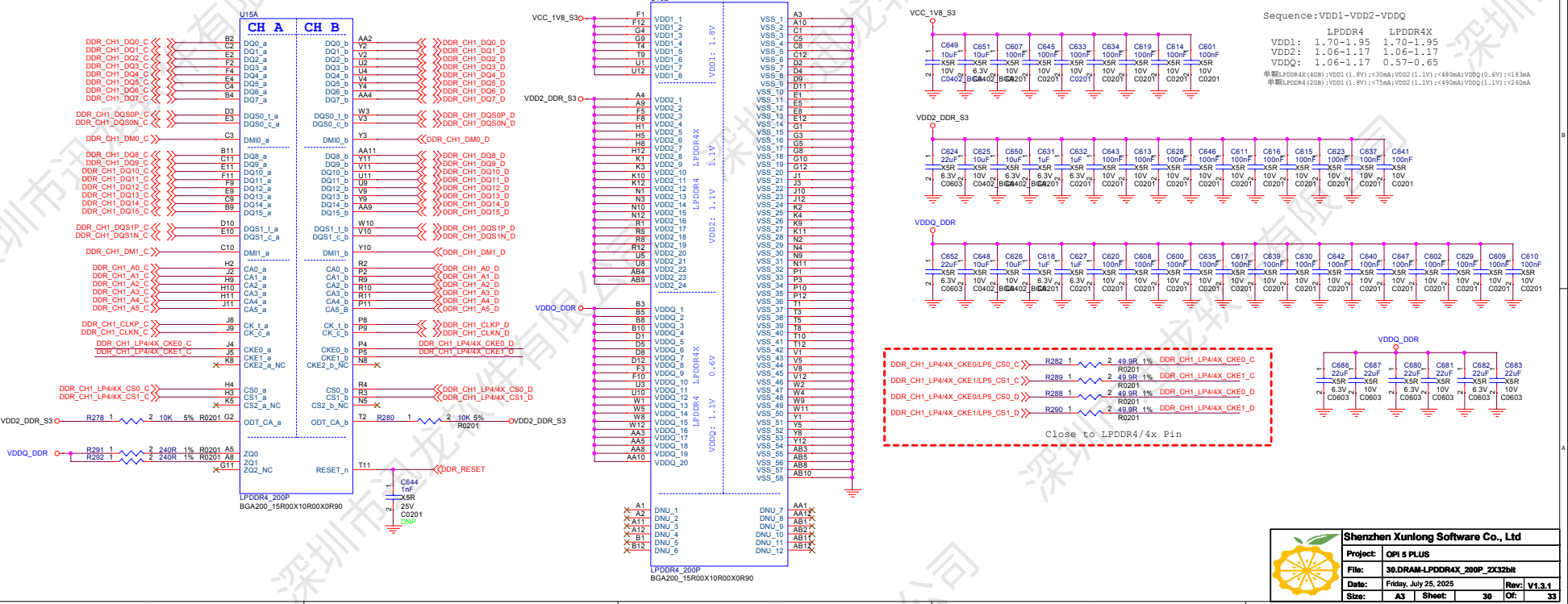
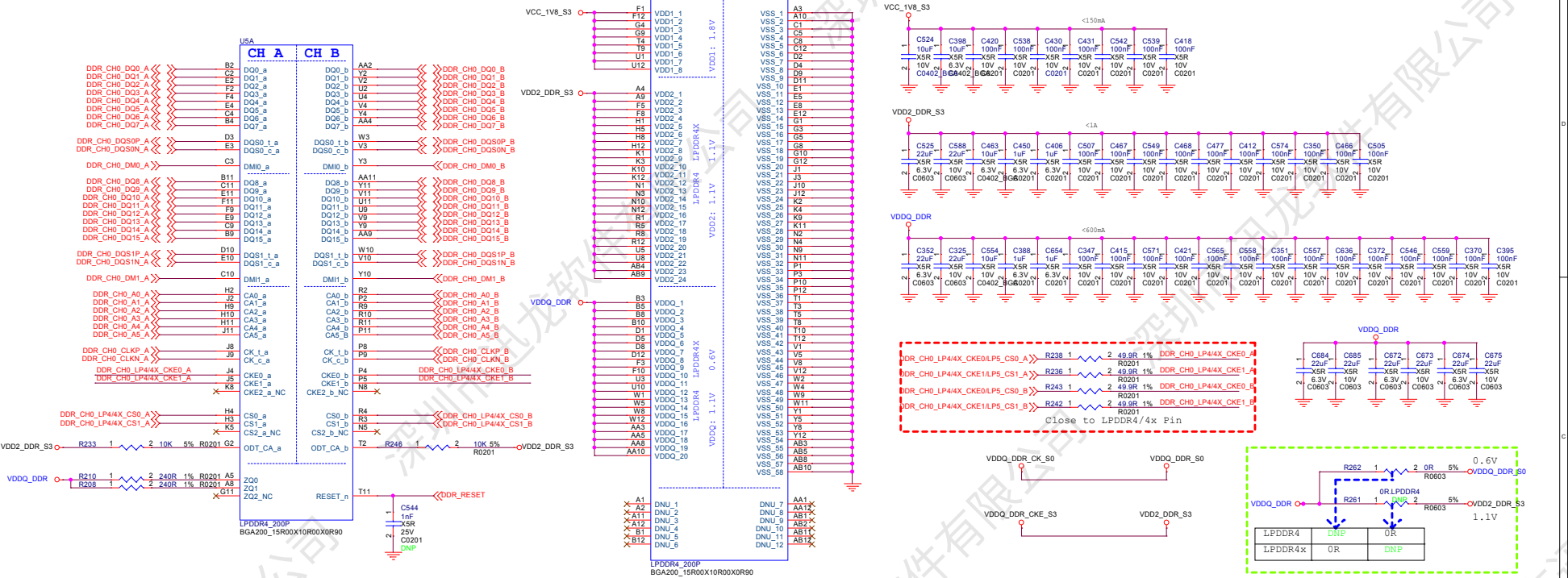
CTIA (L, R, G, M)

Analog MIC



Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	29.Audio Codec			
Date:	Friday, July 25, 2025			Rev: V1.3.1
Size:	A3	Sheet:	29	Of: 33

LPDDR4/4X





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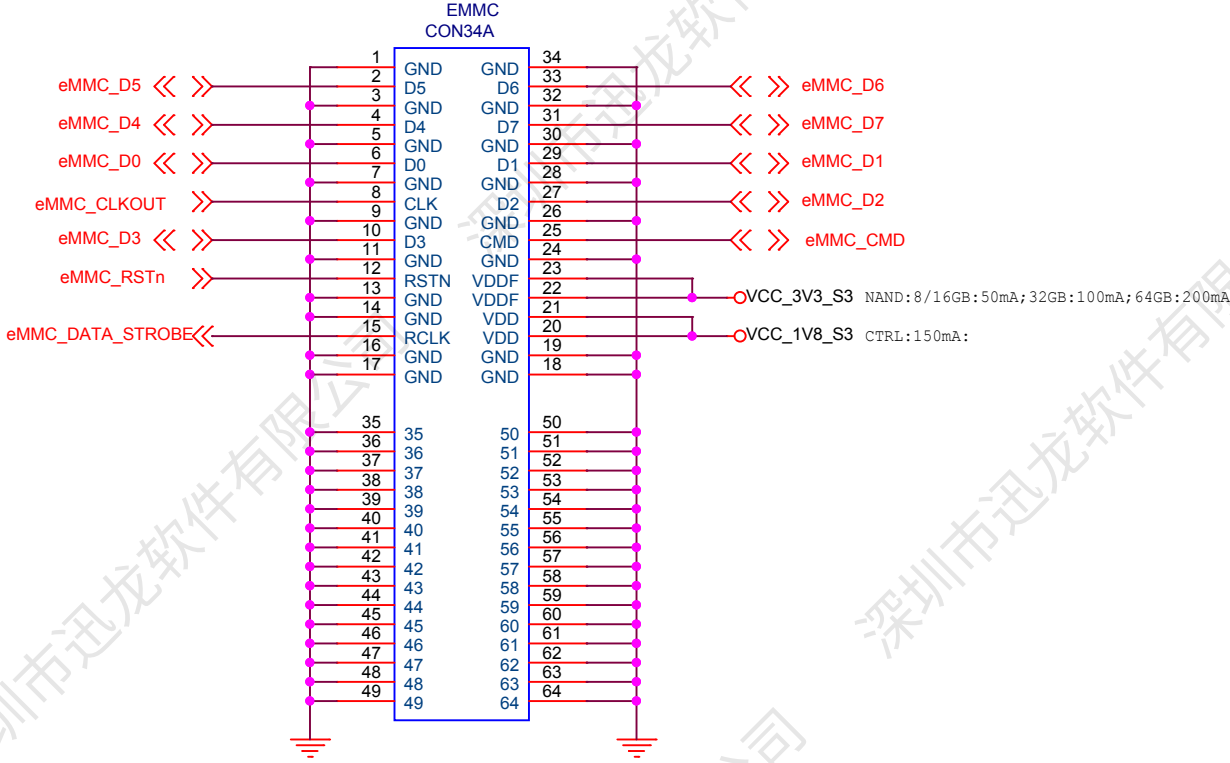
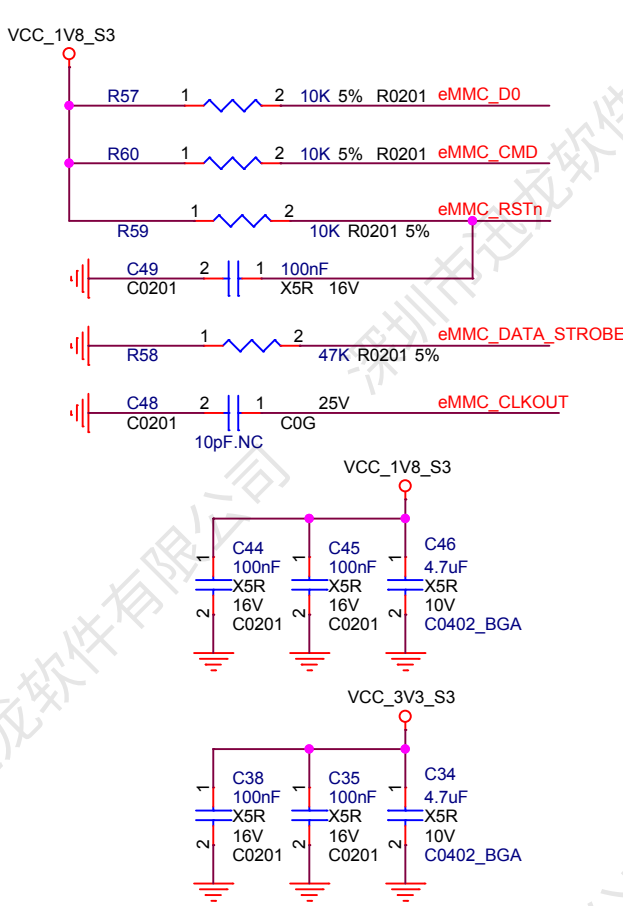
Project: **OP15 PLUS**

File: **3D.DRAM-LPDDR4X_200P_2X32bit**

Date: **Friday, July 25, 2025**

Size: **A3** Sheet: **30** Of: **33**

eMMC FLASH

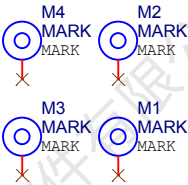
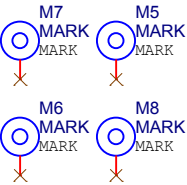


Shenzhen Xunlong Software Co., Ltd				
Project:	OPI 5 PLUS			
File:	31.Flash-eMMC Flash			
Date:	Friday, July 25, 2025	Rev:	V1.3.1	
Size:	A3	Sheet:	31	Of: 33

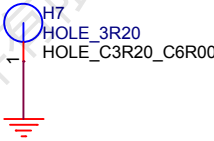
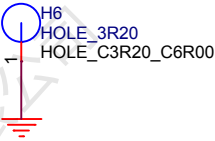
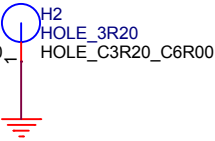
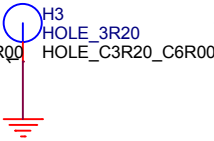
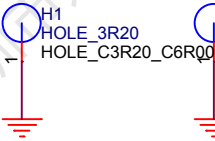
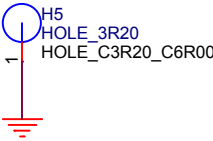
Mark

TOP Mark

BOTTOM Mark



HOLE



HEATSINK

	Shenzhen Xunlong Software Co., Ltd				
	Project:	OPI 5 PLUS			
	File:	33.Mark/Hole/Heatsink			
	Date:	Friday, July 25, 2025			Rev: V1.3.1
	Size:	A3	Sheet:	33	Of: 33